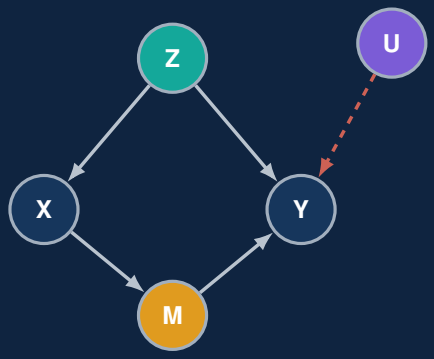


ADVANCED ARTIFICIAL INTELLIGENCE

Probabilistic Graphical Models and Causal Inference

Session 12

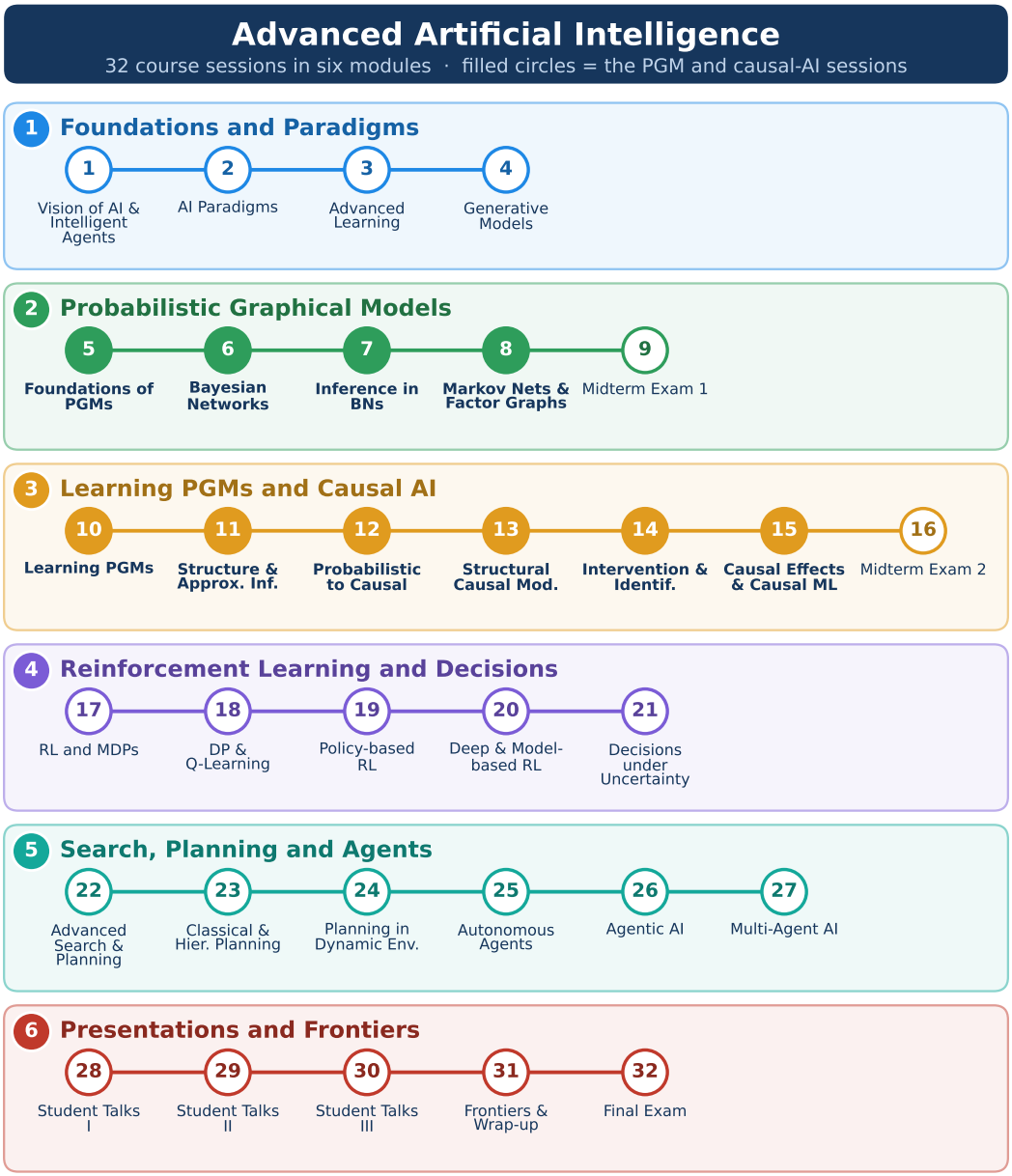
Correlation vs. Causation: The Conceptual Pivot

Lecture Notes

Instructor: Prof. M. Arashi

About these notes

These notes cover Session 12 of the course *Advanced Artificial Intelligence*, part of its probabilistic-graphical-model and causal-AI block (course sessions 5–15). The session follows a fixed structure: a roadmap and learning objectives, a "you are here" course map, the core theory organized in parts, a dedicated part on mathematical statistics, fully worked numerical and analytical examples, and a part connecting the topic to intelligent agents, agentic AI, reinforcement learning, multi-agent systems and the other sessions of the course. Every number is reproduced by the companion Python scripts (`python/s12_examples.py` for the numbers, `python/s12_figures.py` for the figures), and `questions/s12_questions.tex` contains ten questions with complete answers.



The Advanced Artificial Intelligence course. Filled circles mark the PGM and causal-AI sessions.

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SESSION 12 ADVANCED ARTIFICIAL INTELLIGENCE

Correlation vs. Causation: The Conceptual Pivot

Correlation versus causation, causal questions, causal graphs, confounding and selection bias

Advanced Artificial Intelligence · Probabilistic Graphical Models and Causal Inference

Instructor: Prof. M. Arashi

PART I
Two Real Reversals
Berkeley, kidney stones, batting averages: Simpson's paradox with real data.

PART II
Causal Questions & Graphs
The ladder of causation, kinds of causal question, what an arrow claims.

PART III
Confounding & Selection
Back-door paths, randomization, colliders, Berkson, survivorship, M-bias.

PART IV
Mathematical Statistics
Reversal conditions, collider bias, collapsibility, omitted variables, randomization.

PART V
AI Connections
Agent logs, offline RL, leaderboards, shortcut learning, fairness.

LEARNING OBJECTIVES

- Reproduce the Berkeley, kidney-stone and Justice–Jeter reversals from the raw counts and explain each one with a causal graph.
- Distinguish the three rungs of the ladder of causation and classify a question as associational, interventional or counterfactual.
- State precisely what a causal DAG claims beyond a Bayesian network, and recognise confounders, mediators and colliders.
- Explain confounding and selection bias as two different graphical mechanisms, and say why "adjust for everything" fails.
- Prove when Simpson's reversal can and cannot occur, compute collider bias exactly for Gaussian selection, and derive the omitted-variable bias formula.
- Show that randomization makes $T \perp\!\!\!\perp (Y(0), Y(1))$ and hence identifies the average treatment effect.
- Diagnose confounding and selection in logged data of LLM agents, offline RL and multi-agent leaderboards.

Where we are. Course sessions 5–11 built the toolkit of probabilistic graphical models — conditional independence, Bayesian and Markov networks, inference, learning — all describing

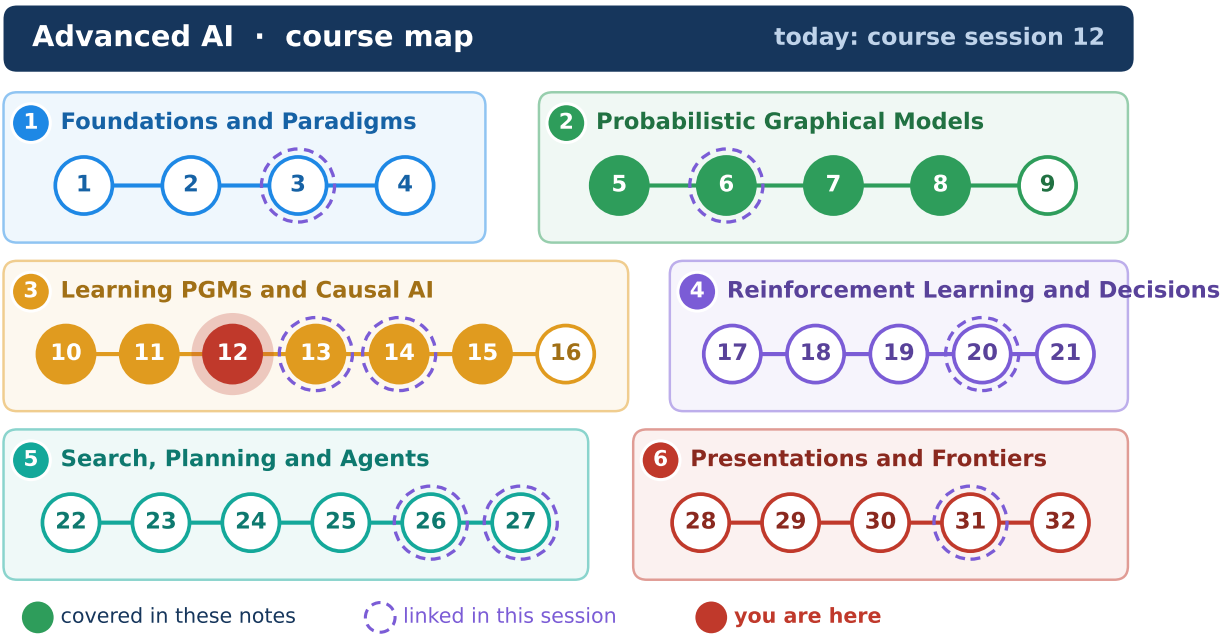


Figure 12.1. Where this session sits in the Advanced AI course. Dashed purple rings mark the course sessions linked today.

a *single, fixed* joint distribution P. This session (course session 12) is the hinge of the causal block: it shows with real data that P alone cannot answer "what if we act?" questions. Structural causal models (Session 13), the do-calculus (Session 14) and causal effect estimation (Session 15) build the remedy; reinforcement learning (course sessions 17–20), agentic AI (course session 26) and multi-agent AI (course session 27) then rely on today's distinction: an agent's actions are interventions, not observations.

OPENING MOTIVATION

Every tool built in Sessions 5–11 — CI, DAGs, variable elimination, Markov networks, EM, MCMC — computes properties of a single fixed probability distribution P, generated by observing the world as it already is. Today we show, with real published data, that this entire toolkit can give you the *wrong sign* of an effect if you ask it the wrong question. Not a slightly-off estimate: the wrong sign, in data (such as the kidney-stone study below) where the comparison is used to choose a patient's treatment. This is not a footnote or a caveat to append to Sessions 5–11; it is the reason a second, distinct mathematical formalism (Sessions 13–15) is necessary at all.

Healthcare

Which treatment should a new patient receive? Observational records reflect how doctors *chose* treatments, not what the treatments *do*.

Hiring and fairness

Is a screening process biased? Pooled admission or hiring rates mix the effect of the process with who applies where (Berkeley).

LLM agents

Does calling a retrieval tool help? Logs record when an old policy chose to call it — typically on harder queries — and which runs were kept.

PART I

Correlation vs. Causation

Three real data sets in which the pooled association has the wrong sign

PART I · REVERSALS

12.1 Two Real Reversals

Worked Example 12.1 (UC Berkeley graduate admissions, 1973)**.** In fall 1973, UC Berkeley's graduate admissions showed a troubling aggregate pattern: of 8442 male applicants about 44% were admitted, against about 35% of 4321 female applicants (Bickel, Hammel & O'Connell, *Science*, 1975). Restricting to the six largest departments (2691 men, 1835 women; full counts in Table 12.1) the gap is even wider: $1198/2691 = 44.5\%$ of men against $557/1835 = 30.4\%$ of women were admitted, a gap of 14.2 percentage points (Fig. 12.2, left). This looks like exactly the kind of evidence a sex-discrimination lawsuit would want. But when the data are broken down *by department* (middle panel and Table 12.1), something remarkable appears: in four of the six departments (A, B, D, F), women were admitted at an equal or *higher* rate than men. Department A alone admitted $89/108 = 82.4\%$ of its (relatively few) female applicants versus $512/825 = 62.1\%$ of men.

Table 12.1. UC Berkeley 1973, the six largest departments (real data; Bickel, Hammel & O'Connell, 1975). Rates are admitted/applied. n_d = all applicants to department d, $N = \sum_d n_d = 4526$; the column n_d/N is the pooled department mix used for standardization.

Dept	Men: admitted/applied	Women: admitted/applied	n_d	n_d/N
A	$512/825 = 0.621$	$89/108 = 0.824$	933	0.206
B	$353/560 = 0.630$	$17/25 = 0.680$	585	0.129
C	$120/325 = 0.369$	$202/593 = 0.341$	918	0.203
D	$138/417 = 0.331$	$131/375 = 0.349$	792	0.175
E	$53/191 = 0.277$	$94/393 = 0.239$	584	0.129
F	$22/373 = 0.059$	$24/341 = 0.070$	714	0.158
All six	$1198/2691 = 0.445$	$557/1835 = 0.304$	4526	1

The resolution: women disproportionately applied to competitive departments with low admission rates for everyone (Department C admits $(120 + 202)/(325 + 593) = 322/918 = 35.1\%$ of all its applicants), while men disproportionately applied to less competitive departments (Department A admits $(512 + 89)/(825 + 108) = 601/933 = 64.4\%$ of all its applicants). In numbers: of the 2691 men, $825 + 560 = 1385$ applied to the two easiest departments A and B, i.e. $1385/2691 = 51.5\%$; of the 1835 women only $108 + 25 = 133$ did, i.e. $133/1835 = 7.2\%$. Each sex's pooled rate is an average of its department rates weighted by *its own* department mix, and these mixes are very different; aggregating across departments therefore mixes "where people applied" into what looks like a sex effect at the admission stage. Note the causal direction: department choice is not a cause of sex but a *consequence* of it (sex → department → admission), a point we return to in Section 12.2.

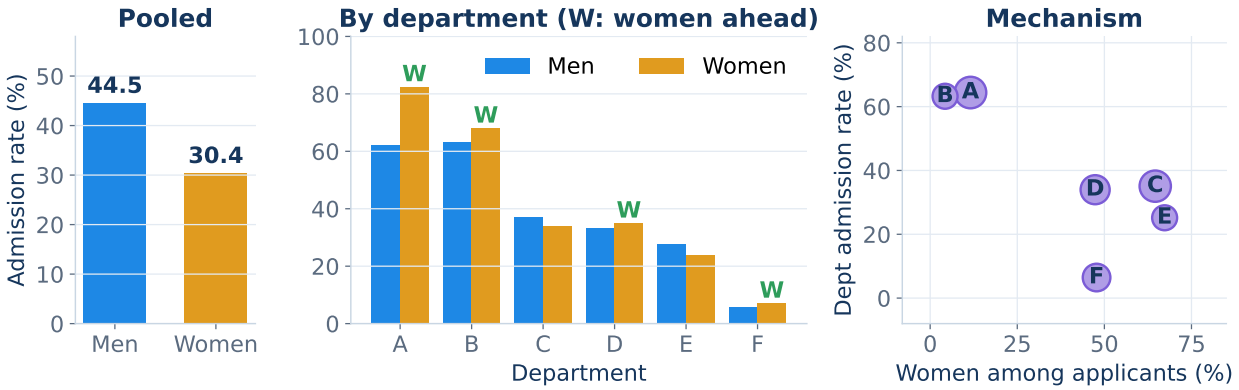


Figure 12.2. Pooled (left) vs. department-stratified (middle) admission rates, UC Berkeley 1973, six largest departments; "W" marks departments where women's rate is at least men's. Right: each department's overall admission rate against the share of women among its applicants (bubble area $\propto$ applicants). Real data (Bickel, Hammel & O'Connell, 1975).

If both sexes faced the *same* department mix — the pooled mix n_d/N of all 4526 applicants (last column of Table 12.1) — their department-specific rates would give (*direct standardization*, Section 12.9)

$$\begin{aligned}\text{men: } & 0.206(0.621) + 0.129(0.630) + 0.203(0.369) + 0.175(0.331) + 0.129(0.277) + 0.158(0.059) \\ & = 0.1279 + 0.0815 + 0.0749 + 0.0579 + 0.0358 + 0.0093 = 0.387, \\ \text{women: } & 0.206(0.824) + 0.129(0.680) + 0.203(0.341) + 0.175(0.349) + 0.129(0.239) + 0.158(0.070) \\ & = 0.1699 + 0.0879 + 0.0691 + 0.0611 + 0.0309 + 0.0111 = 0.430,\end{aligned}$$

i.e. 38.7% for men and 43.0% for women (the products are computed from unrounded rates).

PYTHON LAB · `s12_examples.py` · `ex_berkeley()`

```
aggregate:  men 1198/2691 = 0.445, women 557/1835 = 0.304
Men applying to A or B: 1385/2691 = 0.515; women:  133/1835 = 0.072
terms (n_d/N)*rate, men:  0.1279 + 0.0815 + 0.0749 + 0.0579 + 0.0358 +
0.0093
terms (n_d/N)*rate, women:  0.1699 + 0.0879 + 0.0691 + 0.0611 + 0.0309 +
0.0111
Standardised to pooled department mix:  men 0.387, women 0.430
```

Worked Example 12.2 (Kidney stone treatment, Charig et al. 1986)**.** A comparative study of two kidney-stone treatments — open surgery (Treatment A) and percutaneous nephrolithotomy (Treatment B) — reported, in aggregate, that Treatment B succeeded 82.6% of the time ($234 + 55 = 289$ successes among $270 + 80 = 350$ patients) versus 78.0% ($81 + 192 = 273$ among $87 + 263 = 350$) for Treatment A (Fig. 12.3, left), suggesting B is the better treatment. Yet stratified by stone size (Fig. 12.3, middle, using the standard tabulation by Julious & Mullee, *BMJ*, 1994): for *small* stones, A succeeds 93.1% (81/87) of the time versus B's 86.7% (234/270); for *large* stones, A succeeds 73.0% (192/263) versus B's 68.8% (55/80). Treatment A is **better in both subgroups**, yet worse overall. This is not a rounding artifact — it is a full reversal of direction.

The mechanism is identical to Berkeley's: physicians preferentially used the less invasive Treatment B on *small* (easier) stones and reserved open surgery (Treatment A) for *large*, harder cases (Fig. 12.3,

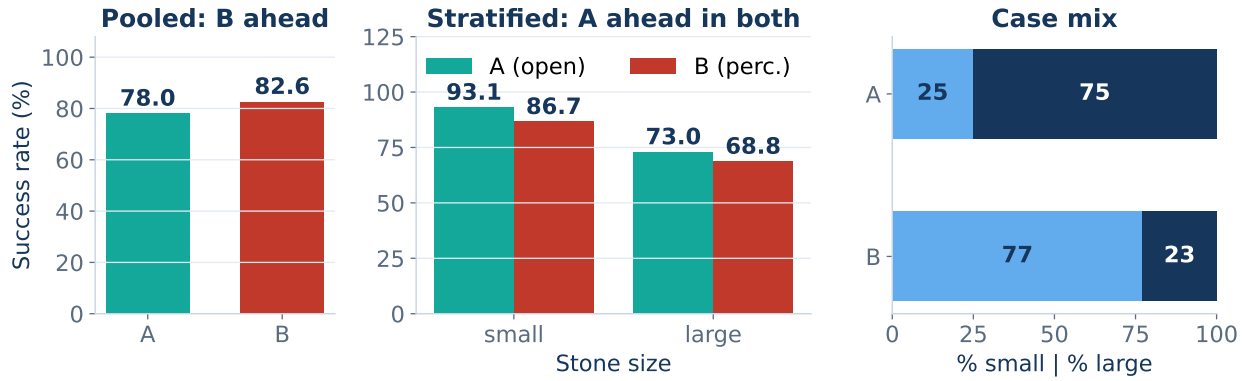


Figure 12.3. Kidney stones. Pooled (left) vs. stone-size-stratified (middle) success rates, and the case mix of each treatment (right): $263/350 = 75\%$ of A's patients had large stones, against $80/350 = 23\%$ of B's. Real data (Charig et al., 1986).

right). Stone size drives both which treatment a patient received and how likely they were to recover regardless of treatment. Aggregating over stone size mixes "how severe was your case" into what looks like a treatment effect.

PYTHON LAB · s12_examples.py · ex_kidney()

```
small stones:  A 81/87 = 0.9310; B 234/270 = 0.8667
large stones:  A 192/263 = 0.7300; B 55/80 = 0.6875
aggregate A: 273/350 = 0.7800; aggregate B: 289/350 = 0.8257
```

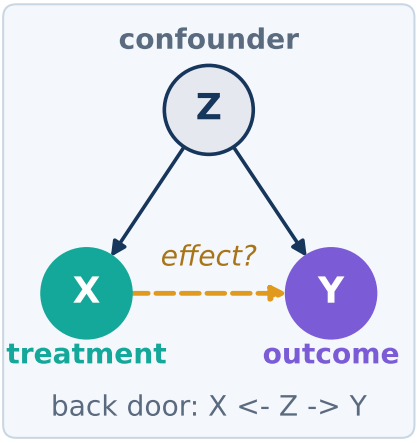
Definition 12.1 (Simpson's paradox)**.** An association between X and Y that holds in the aggregated population reverses (or vanishes) when the population is stratified by a third variable Z.

PART I · REVERSALS

12.2 The Graphical Diagnosis

The kidney-stone study has the causal structure of Fig. 12.4: stone size Z is a common cause of the treatment X (physicians chose by stone size) and of the outcome Y (recovery). In Session 6's vocabulary, Z is the middle node of a *fork* $X \leftarrow Z \rightarrow Y$. Marginally, this fork *induces* an association between X and Y that has nothing to do with any effect of X on Y — exactly as the latent "enthusiast" class of Session 5 made two unrelated film ratings dependent. The difference now is practical: the spurious dependence sits between the treatment we must choose and the outcome we care about. The batting-average and marketing examples of Section 12.14 and the agent logs of Worked Example 12.16 have the same fork structure.

Berkeley shares the *arithmetic* of the reversal (the two groups have different distributions over Z) but not the graph: sex cannot be caused by department, so the structure is $X \rightarrow Z \rightarrow Y$ (sex influences department choice, which influences admission), possibly with a direct arrow $X \rightarrow Y$. Department is a *mediator*, not a confounder. Stratifying by department then answers a different, narrower question — is there a *direct* effect of sex on admission within the department that made the decision? — whereas the pooled comparison measures the total association. Which of the two measures "discrimination" is a causal and normative question, not a statistical one (Pearl 2009, Sec. 6.1; see the discussion question in Section 12.16).



Example	Z	X	Y
Kidney stones	stone size	A or B	success
Agent logs	difficulty	tool call	success
Justice/Jeter	season	player	hit
Ad (hypoth.)	intent	ad shown	purchase

Figure 12.4. The confounding fork: Z drives both the "treatment" X and the outcome Y, opening the back-door path $X \leftarrow Z \rightarrow Y$. Right: the examples of this session with this structure (kidney stones and batting averages are real data; the agent logs of Worked Example 12.16 and the advertisement example of Worked Example 12.12 are hypothetical). Berkeley is *not* in the list: there department is a mediator (see text).

Definition 12.2 (Confounder, informally)**.** A variable Z is a *confounder* for the relationship between X and Y if Z causally influences both X and Y. Naively comparing $P(Y \mid X = 1)$ to $P(Y \mid X = 0)$ then reflects a mixture of (i) any genuine effect of X on Y and (ii) the spurious association induced by Z.

The seemingly obvious fix — "stratify by (control for) Z" — is exactly what resolved the kidney-stone reversal, and it will be formalized properly as the *back-door criterion* in Session 14. Berkeley already shows that the fix needs a graph: stratifying by a mediator changes the question being answered. But notice we have already seen, in Session 6, that conditioning is not *always* the right move: conditioning on a **collider** ($X \rightarrow Z \leftarrow Y$, rather than a confounding fork $X \leftarrow Z \rightarrow Y$) *creates* spurious association where none existed before. A blanket rule of "control for everything you measured" is therefore not merely inelegant — it can be actively wrong, and can manufacture exactly the kind of paradox we've just resolved, in the opposite direction. This tension — when to condition, when not to — is precisely why Session 14 will insist on a checkable graphical *criterion* rather than a rule of thumb.

PART I · REVERSALS

12.3 Why $P(Y \mid X{=}x)$ Is the Wrong Quantity

Here is the sharpest way to state what went wrong in the kidney-stone example. The number we computed, $P(Y = 1 \mid X = x)$, answers: *"among the units we observed that happened to have $X = x$, what fraction had $Y = 1$?"* It is a fact about who ended up in each group — and who ended up in each treatment group was itself driven by stone size Z. What we actually want to know is different: *"if we were to take a patient and force X to equal x, overriding whatever process would ordinarily determine X for them, what would happen to Y?"* This is not a question about a subpopulation defined by an observed value of X; it is a question about an *action* performed on a unit, irrespective of how that unit would ordinarily have arrived at that value of X.

We introduce notation for this now, informally, and formalize it fully next session:

$$\underbrace{P(y \mid x)}_{\text{observational: who has } X{=}x?} \qquad \text{vs.} \qquad \underbrace{P\big(y \mid \mathrm{do}(x)\big)}_{\text{interventional: set } X{=}x \text{ for everyone}} . \tag{12.1}$$

For the fork of Fig. 12.4, intervening on X cuts the arrow $Z \to X$ but leaves the distribution of Z untouched, which gives (Session 13 derives this from the truncated factorization of a causal model; Session 14 generalizes it as the back-door adjustment formula)

$$P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid x, z)\, P(z) \qquad \text{whereas} \qquad P(y \mid x) = \sum_z P(y \mid x, z)\, P(z \mid x). \tag{12.2}$$

The two differ only in the weights: $P(z)$ versus $P(z \mid x)$. Every reversal in this session lives in that difference.

Key Takeaway 12.1. In general, $P(y \mid x) \neq P(y \mid \mathrm{do}(x))$. Every session so far has computed the left-hand side. Nothing in Sessions 5–11 — no amount of more data, better inference algorithms, or more careful sampling — closes this gap, because the gap is not a statistical estimation problem. It is a conceptual one: the graphs of Sessions 5–11 (Bayesian networks and Markov networks alike) encoded only a factorization of P, and a factorization alone does not specify what happens when a piece of the generative process is overridden rather than observed.

12.3.1 Six reasons two variables can be correlated

Confounding is only one way to obtain an association without a causal effect. Before reading $\mathrm{Cov}(X, Y) \neq 0$ as "X causes Y", six explanations must be weighed: (1) causation $X \to Y$; (2) reverse causation $Y \to X$; (3) confounding $X \leftarrow Z \to Y$; (4) selection on a common effect, $X \to S \leftarrow Y$ with only $S = 1$ observed; (5) chance (small samples, many comparisons); (6) measurement error, e.g. adjusting for a noisy proxy of Z.

Definition 12.3 (Reichenbach's common-cause principle)**.** If two variables X and Y are statistically dependent, then either X causes Y, or Y causes X, or there is a third variable Z that causes both; in the last case Z *screens off* the dependence: $X \perp\!\!\!\perp Y \mid Z$. (Reichenbach 1956; see Peters, Janzing & Schölkopf 2017, Ch. 1.)

The principle is the bridge between Sessions 5–11 and causality: dependence demands a causal explanation. Two caveats sharpen it. First, it presupposes that we look at the *whole* population; a dependence manufactured by *selection* (reason 4) has no common cause at all — the sample was filtered on a common *effect*. Second, *reverse causation* (reason 2) is indistinguishable from forward causation by $P(x, y)$ alone, as the next example shows.

Worked Example 12.3 (Reverse causation is invisible in the joint distribution)**.** *(Analytical.)* Model 1: $X \sim N(0,1)$, $Y := 0.6X + 0.8U_Y$. Model 2: $Y \sim N(0,1)$, $X := 0.6Y + 0.8U_X$, with independent standard normal noises U_X, U_Y, independent of the root variable. In Model 1, $\mathrm{Var}X = 1$, $\mathrm{Var}Y = 0.6^2\mathrm{Var}X + 0.8^2\mathrm{Var}U_Y = 0.36 + 0.64 = 1$ and $\mathrm{Cov}(X, Y) = \mathrm{Cov}(X, 0.6X + 0.8U_Y) = 0.6\mathrm{Var}X = 0.6$; Model 2 gives the same three numbers with the roles of X and Y swapped. Both pairs are jointly Gaussian with mean 0, so both models induce the *same* bivariate normal law $N\big(0, \left(\begin{smallmatrix} 1 & 0.6 \\ 0.6 & 1 \end{smallmatrix}\right)\big)$. Yet in Model 1 setting $X = 1$ gives $Y = 0.6 \cdot 1 + 0.8U_Y$, so $\mathbb{E}[Y \mid \mathrm{do}(X = 1)] = 0.6$, while in Model 2 the equation $Y \sim N(0,1)$

does not involve X, so $\mathbb{E}[Y \mid \mathrm{do}(X = 1)] = \mathbb{E}[Y] = 0$: setting X does nothing to its own cause. No amount of observational data distinguishes them; the difference lives on rung 2 of the ladder (Section 12.4).

PYTHON LAB · s12_examples.py · ex_reverse()

```
simulation, n = 200000 draws per model, seed 7 (sample covariance matrices):
X->Y: cov = [[0.998, 0.599], [0.599, 0.997]]; Y->X: cov = [[0.999, 0.601],
[0.601, 1.002]]
```

PART II

Causal Questions and Causal Graphs

What we ask, what an arrow claims, and which variables play which role

PART II · CAUSAL QUESTIONS

12.4 Pearl's Ladder of Causation

Pearl (2009; Pearl & Mackenzie 2018) organises causal questions into three rungs (Fig. 12.5): *association* (seeing; supervised learning and LLM pre-training live here), *intervention* (doing; experiments, A/B tests and every action of an RL agent) and *counterfactuals* (imagining; attribution, regret and credit assignment). Each rung asks a strictly richer kind of question, and — this is the central message of the ladder — *information from a lower rung is in general insufficient to answer questions on a higher rung.* Worked Example 12.3 already showed two models that agree on all rung-1 queries and disagree on a rung-2 query.

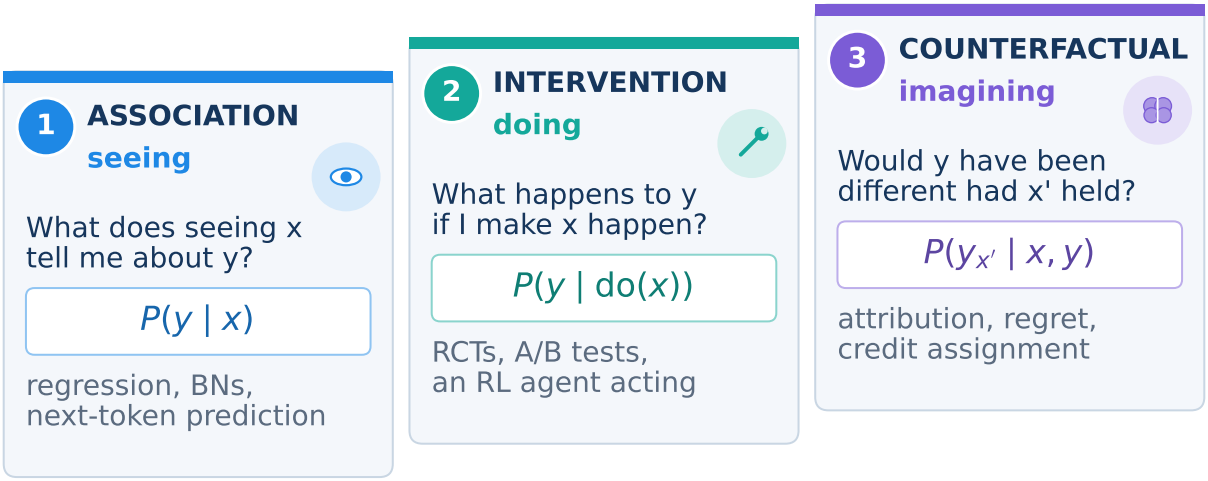


Figure 12.5. The ladder of causation. Rung 1 is the whole of Sessions 5–11; rungs 2 and 3 need a causal model (Sessions 13–15).

PART II · CAUSAL QUESTIONS

12.5 A Taxonomy of Causal Questions

"Does X cause Y?" hides several different questions. Table 12.2 separates them; each will reappear with its own estimand later in the block.

Table 12.2. Kinds of causal question. EoC = effect of a cause, CoE = cause of an effect; $\mathrm{do}(1)$ abbreviates $\mathrm{do}(X = 1)$.

Question type	Example	Target quantity	Treated in
Effect of a cause (EoC)	Does drug A lower mortality?	$\mathrm{ATE} = \mathbb{E}[Y \mid \mathrm{do}(1)] - \mathbb{E}[Y \mid \mathrm{do}(0)]$	Sessions 14–15
Mediation	How much of the effect runs through blood pressure?	direct vs. indirect effect	Sessions 13–14
Cause of an effect (CoE) / attribution	Did the faulty sensor cause the crash?	probability of necessity, $P(Y_{x'} = 0 \mid X = x, Y = 1)$ ($Y_{x'}$: outcome had X been x')	Session 13
Policy question	What if the agent always calls the verifier?	$P(y \mid \mathrm{do}(\pi))$ for a policy π	Session 14; course sessions 17–20

PART II · CAUSAL QUESTIONS

12.6 Causal Graphs: What an Arrow Claims

In Sessions 5–6 (and in the structure learning of Session 11) a DAG was a bookkeeping device for a factorization $P(v) = \prod_i P(v_i \mid \mathrm{pa}_i)$; many DAGs (a whole Markov-equivalence class) represent the same P, and the direction of an arrow carried no meaning beyond that. A *causal* DAG makes a much stronger claim.

Definition 12.4 (Causal DAG)**.** A DAG G over variables V is a *causal DAG* if (i) an arrow $V_j \to V_i$ means that V_j is a *direct cause* of V_i relative to V: some intervention on V_j changes the distribution of V_i while all other variables in V are held fixed by intervention; and (ii) every common cause of two or more variables in V is itself in V (*causal sufficiency*), or is represented explicitly as a latent node.

Definition 12.5 (Causal Markov assumption, preview)**.** Given a causally sufficient causal DAG G, every variable is independent of its non-descendants given its direct causes: $V_i \perp \perp \mathrm{nd}(V_i) \mid \mathrm{pa}(V_i)$. Equivalently, P factorizes over G. Under an intervention $\mathrm{do}(V_k = v_k)$ the factor $P(v_k \mid \mathrm{pa}_k)$ is removed and all other factors remain *unchanged* (modularity).

This modularity is what a Bayesian network does not claim: the conditionals are *autonomous mechanisms*, so an intervention replaces one and leaves the rest intact (derived from structural equations in Session 13). Table 12.3 summarises the difference; note that its assumptions live in the arrows that are *absent*.

Table 12.3. The same picture, two different claims.

	Bayesian network (Sessions 6–11)	Causal DAG (Sessions 12–15)
An arrow $X \to Y$ means	X appears in the factor of Y	X is a direct cause of Y
A missing arrow means	a conditional independence	no direct causal effect (the strongest assumption)
Arrow direction	conventional; Markov-equivalent DAGs are interchangeable	substantive; $X \to Y$ and $Y \to X$ are different models
Answers	rung 1 queries	rung 1 and rung 2 queries (rung 3 with an SCM)
Under $\mathrm{do}(x)$	undefined	delete the factor of X, keep the others

12.6.1 Three roles a third variable can play

When we study the effect of X on Y, a third variable can play one of three basic structural roles (Fig. 12.6): a confounder Z (Definition 12.2), a mediator M or a collider S (defined next). The decision "should I adjust for this variable?" depends on its role, never on how strongly it correlates with X or Y.

Definition 12.6 (Mediator and collider)**.** On a path between X and Y, M is a *mediator* if the path contains $X \to M \to Y$ (M lies on a causal path from X to Y); S is a *collider* on a path if both edges of the path point into it, $\cdots \to S \leftarrow \cdots$.

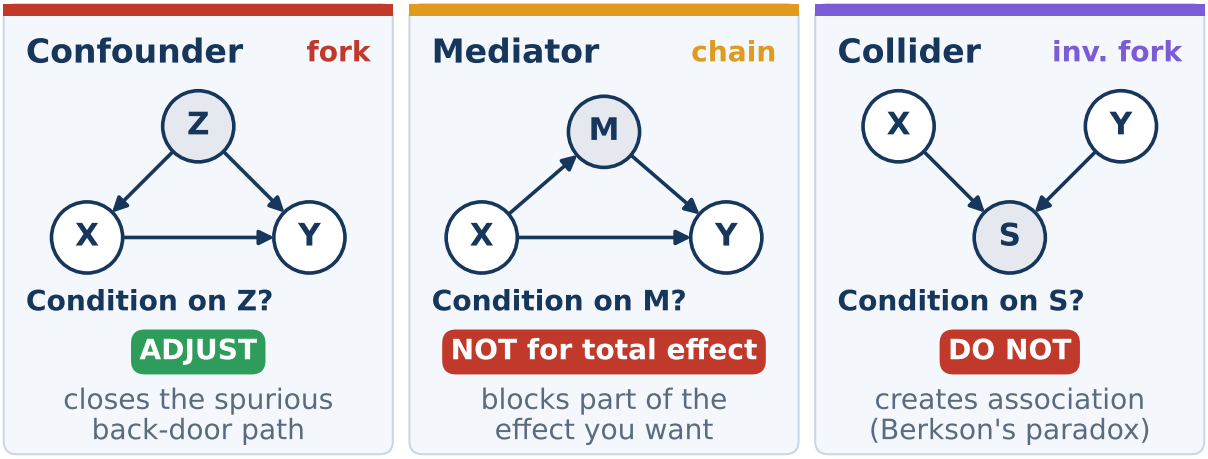


Figure 12.6. Confounder, mediator, collider. For the *total* effect of X on Y: adjust for the confounder, do not adjust for the mediator, never adjust for the collider (or any of its descendants).

PART III

Confounding and Selection Bias

Two different graphical mechanisms that make $P(y \mid x)$ lie

PART III · BIAS

12.7 Confounding

Definition 12.7 (Confounding bias; back-door path)**.** The effect of X on Y is *confounded* if $P(y \mid x) \neq P(y \mid \mathrm{do}(x))$ for some x, y. A *back-door path* from X to Y is a path that starts with an arrow *into* X, such as $X \leftarrow Z \rightarrow Y$. In a causal DAG, open back-door paths are the graphical source of confounding; Session 14 shows that blocking all of them (without opening new ones) by conditioning on a set Z yields Eq. (12.2).

The definition is causal: no property of P alone certifies the absence of confounding, and the traditional checklist ("Z associated with X and with Y, not on the causal path") is not a valid substitute (Hernán & Robins 2020, Ch. 7). Passing the checklist does not make adjustment correct: in the M-bias example (Worked Example 12.9) the pre-treatment variable M is associated with both X and Y and is not on the causal path, yet adjusting for it *creates* bias. Failing to find a variable that passes the checklist does not certify the absence of confounding either: if the common cause is unmeasured (as in Worked Example 12.18), there is no variable to check, and the data alone cannot reveal it.

12.7.1 Randomization: the gold standard

The cleanest way to remove confounding is to make X itself random: if a coin decides the treatment, nothing about the unit can influence X, so every arrow into X, and with it every back-door path, is cut. Section 12.13 states this precisely with potential outcomes: writing T for a randomized binary treatment and $Y(0), Y(1)$ for the outcomes the unit would have under each treatment, randomization gives $T \perp\!\!\!\perp (Y(0), Y(1))$ (Theorem 12.3). RCTs in medicine and A/B tests in technology are the operational form of rung 2: we *do* x instead of watching who has x.

Measurement error and proxies. Often the confounder is measured with error (an income bracket for wealth, a triage score for severity). Adjusting for a noisy proxy $W = Z + e$ removes only part of the bias — *residual confounding* — and the remainder grows as the proxy's reliability falls (Worked Example 12.8).

PART III · BIAS

12.8 Selection Bias

Definition 12.8 (Selection bias)**.** Let $S \in \{0, 1\}$ indicate whether a unit enters the data set. *Selection bias* occurs when the analysed distribution $P(x, y \mid S = 1)$ differs from the target $P(x, y)$ in a way that changes the association of interest; graphically, this happens when S is a descendant of Y (or of a common descendant of X and Y), so that conditioning on $S = 1$ conditions on a collider or on the outcome itself.

Berkson's paradox (collider bias). Berkson (1946) observed that two diseases that are independent in the population appear *negatively* associated among hospital patients, because

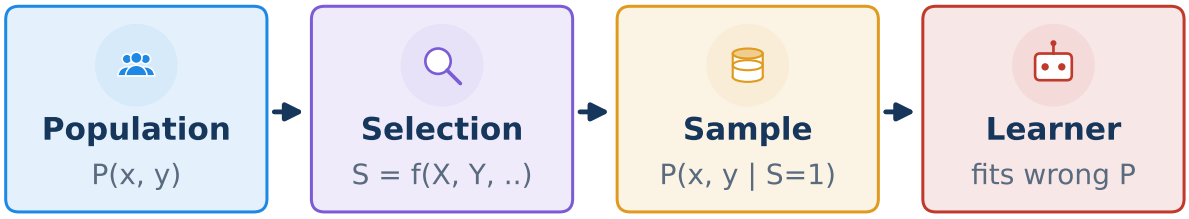


Figure 12.7. The selection-bias pipeline: a population, a selection mechanism that depends on the variables of interest, a filtered sample, and a learner that faithfully fits the wrong distribution.

a person is hospitalised if they have one *or* the other: hospitalisation is a collider. The same mechanism makes talent and effort negatively correlated among admitted students, or attractiveness and kindness among people one has dated. Figure 12.8 simulates it and Theorem 12.2 computes it exactly.

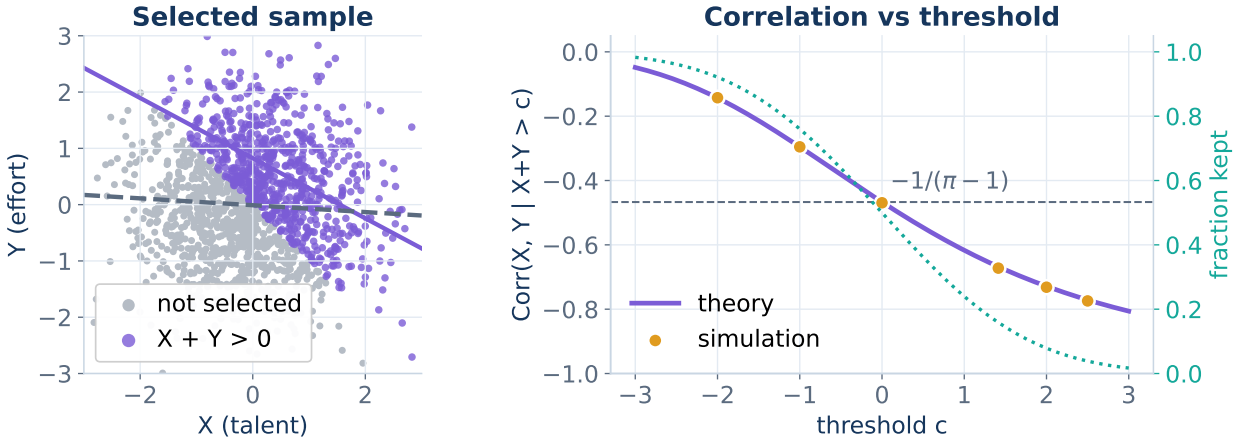


Figure 12.8. Berkson's paradox. Left: 1500 simulated pairs $X, Y \overset{\text{iid}}{\sim} N(0,1)$ (seed 7, `fig_berkson()` in `s12_figures.py`); the purple points are the units kept by the rule $X + Y > 0$, the grey points those discarded. The least-squares line through the kept units (purple, solid) is clearly decreasing, whereas the line through all 1500 points (grey, dashed) is essentially flat. Right: the exact correlation in the selected sample as a function of the threshold c (Theorem 12.2), simulated values at $c \in \{-2, -1, 0, \sqrt{2}, 2, 2.5\}$ (4×10^5 pairs each, Worked Example 12.5), and the fraction of the population kept, $1 - \Phi(c/\sqrt{2})$ (dotted, right axis).

Survivorship bias. Selection on the outcome itself. The classic anecdote is Abraham Wald's analysis of returning WWII bombers: bullet holes were rare on the engines not because engines were never hit but because planes hit there did not return. In AI, data sets of "successful trajectories" used to evaluate or fine-tune agents, or benchmarks that silently drop crashed runs, are survivor samples (Worked Example 12.17).

Self-selection and M-bias. Volunteers, survey respondents and users who choose to rate an answer differ systematically from the population. A subtler trap is *M-bias*: a *pre-treatment* variable M that is a collider between two latent causes, $X \leftarrow U_1 \rightarrow M \leftarrow U_2 \rightarrow Y$. The path is blocked as long as we leave M alone; adjusting for M "because it was measured before treatment" opens it (Worked Example 12.9).

Table 12.4. Confounding vs. selection bias.

	Confounding	Selection bias
Graph	common *cause*: $X \leftarrow Z \rightarrow Y$	common *effect* conditioned on: $X \rightarrow S \leftarrow Y$, or $Y \rightarrow S$
Exists in the full population?	yes	no; created by the sampling/filtering
Fixed by randomizing X?	yes	no (an RCT with selective drop-out is still biased)
Fixed by adjusting?	adjust for Z (back-door)	do *not* adjust for S; need a model of the selection probability $P(S = 1 \mid \cdot)$, e.g. inverse probability weighting (IPW) by $1/P(S = 1 \mid \cdot)$

PART IV

The Mathematics of Reversal and Bias

When reversals can happen, how large collider bias is, and why randomization works

12.9 Simpson's Reversal as a Weighted-Average Phenomenon

Write $p_{xz} = P(Y = 1 \mid X = x, Z = z)$, $w_z^x = P(Z = z \mid X = x)$ and $p_x = P(Y = 1 \mid X = x) = \sum_z w_z^x p_{xz}$ (law of total probability). The pooled rate of each group is a weighted average of its stratum rates, *with group-specific weights*. Everything follows from that.

Theorem 12.1 (Decomposition of the pooled difference)**.** *For any discrete Z,*

$$p_1 - p_0 = \underbrace{\sum_z w_z^1 \,(p_{1z} - p_{0z})}_{\text{within-stratum effect}} + \underbrace{\sum_z (w_z^1 - w_z^0)\, p_{0z}}_{\text{composition (confounding) term}} . \tag{12.3}$$

If Z is binary, the composition term equals $(w_1^1 - w_1^0)(p_{01} - p_{00})$: the product of the X–Z association and the Z–Y association among the untreated.

Proof. $p_1 - p_0 = \sum_z w_z^1 p_{1z} - \sum_z w_z^0 p_{0z}$. Add and subtract $\sum_z w_z^1 p_{0z}$ and regroup to obtain Eq. (12.3). For binary Z, $\sum_z (w_z^1 - w_z^0) = 0$ gives $w_0^1 - w_0^0 = -(w_1^1 - w_1^0)$, so the composition term is $(w_1^1 - w_1^0)p_{01} - (w_1^1 - w_1^0)p_{00}$. □

Proposition 12.1 (No reversal without X–Z dependence)**.** *If $Z \perp\!\!\!\perp X$, then $p_1 - p_0 = \sum_z P(z)(p_{1z} - p_{0z})$. Hence if $p_{1z} \geq p_{0z}$ for every z, then $p_1 \geq p_0$, with strict inequality if some stratum with $P(z) > 0$ is strict. Consequently a Simpson reversal with binary Z requires both*

$w_1^1 \neq w_1^0$ and $p_{01} \neq p_{00}$, and more precisely

$$(w_1^1 - w_1^0)(p_{01} - p_{00}) < -\sum_z w_z^1 (p_{1z} - p_{0z}) < 0.$$

Proof. If $Z \perp\!\!\!\perp X$, then $w_z^1 = w_z^0 = P(z)$, the composition term vanishes and the pooled difference is a convex combination of the stratum differences, which preserves sign. For the second statement, a reversal means every $p_{1z} - p_{0z} > 0$ (so the within term is positive) while $p_1 - p_0 < 0$; by Theorem 12.1 the composition term must be negative and exceed the within term in absolute value. □

Worked Example 12.4 (The decomposition on the kidney-stone data)**.** Apply Theorem 12.1 with $x = 1$ for Treatment A, $x = 0$ for Treatment B, $Z = 0$ for small and $Z = 1$ for large stones. The data of Example 12.2 give:

1. *Stratum rates.* $p_{10} = 81/87 = 0.9310$, $p_{00} = 234/270 = 0.8667$; $p_{11} = 192/263 = 0.7300$, $p_{01} = 55/80 = 0.6875$. Differences: $p_{10} - p_{00} = 0.0644$, $p_{11} - p_{01} = 0.0425$.
2. *Case-mix weights.* $w_0^1 = 87/350 = 0.2486$, $w_1^1 = 263/350 = 0.7514$; $w_0^0 = 270/350 = 0.7714$, $w_1^0 = 80/350 = 0.2286$.
3. *Within-stratum effect.* $0.2486(0.0644) + 0.7514(0.0425) = 0.0160 + 0.0320 = +0.0480$.
4. *Composition term* (binary form of Theorem 12.1). $(w_1^1 - w_1^0)(p_{01} - p_{00}) = (0.7514 - 0.2286)(0.6875 - 0.8667) = 0.5229 \cdot (-0.1792) = -0.0937$.
5. *Total.* $0.0480 - 0.0937 = -0.0457$, which is indeed $p_1 - p_0 = 0.7800 - 0.8257$.

Treatment A is about 4.8 points better within strata, but its case mix (three quarters large stones) costs it 9.4 points. Re-weighting both arms to the pooled mix of all 700 patients, $P(\text{small}) = (87 + 270)/700 = 357/700 = 0.51$ and $P(\text{large}) = (263 + 80)/700 = 343/700 = 0.49$ (direct standardization, i.e. the left formula of Eq. (12.2)), gives

$$\begin{aligned} \text{A: } & 0.51(0.9310) + 0.49(0.7300) = 0.4748 + 0.3577 = 0.8325, \\ \text{B: } & 0.51(0.8667) + 0.49(0.6875) = 0.4420 + 0.3369 = 0.7789, \end{aligned}$$

a standardized advantage of $0.8325 - 0.7789 = +0.0537$ for A (computed from unrounded rates).

PYTHON LAB · `s12_examples.py` · `ex_simpson_decomposition()`

```
differences A-B = (0.0644, 0.0425); within-stratum products = (0.0160,
0.0320)
w^A_large - w^B_large = 0.5229; pB_large - pB_small = -0.1792
aggregate gap -0.0457 = within-stratum +0.0480 + composition -0.0937
standardised success:  A 0.8325, B 0.7789, difference +0.0537
```

MATHEMATICAL STATISTICS · Standardization and the sure-thing principle

Define the standardized rate $\bar{p}_x = \sum_z P(z)\, p_{xz}$. Because the weights $P(z)$ do not depend on x, Proposition 12.1 applies verbatim: if $x = 1$ is better in every stratum, it is better after standardization. This is Savage's *sure-thing principle*, and Pearl (2009, Sec. 6.1) observes that it holds for $P(y \mid \text{do}(x))$ — provided Z is not affected by X — precisely because of

Eq. (12.2). The paradox disappears once we ask the rung-2 question: the observational conditional p_x violates the sure-thing principle, the interventional one does not. A geometric picture is the *Baker–Kramer plot* (Fig. 12.9): each group's pooled rate lies on the segment joining its two stratum rates, at the abscissa given by its own case mix.

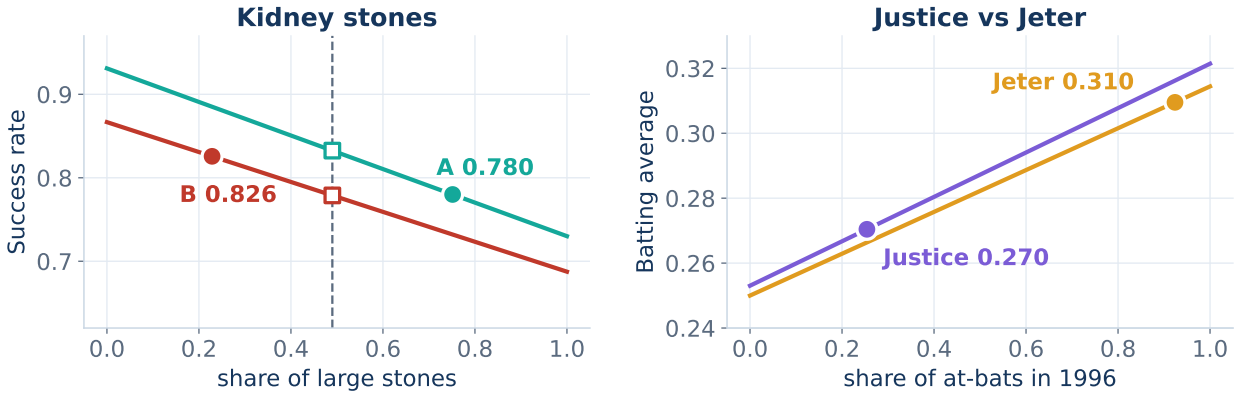


Figure 12.9. Baker–Kramer plots. Each line joins a group's two stratum rates (value at abscissa 0: first stratum; at 1: second stratum); the dot is the pooled rate, located at the group's own mix. Left: kidney stones, abscissa = share of large stones; A's line is higher everywhere, yet A's dot (0.780, at 0.75) is lower than B's (0.826, at 0.23); the squares on the dashed line are the standardized rates 0.8325 and 0.7789 at the pooled mix $P(\text{large}) = 0.49$ (Worked Example 12.4). Right: Justice vs. Jeter, 1995–96, abscissa = share of at-bats in 1996 (Worked Example 12.14). Real data.

MATHEMATICAL STATISTICS · Is it noise? Testing the pooled and the stratified gap

Pooled test. For Berkeley's six departments (Table 12.1) let $\hat{p}_1 = 1198/2691 = 0.4452$ (men, $n_1 = 2691$), $\hat{p}_0 = 557/1835 = 0.3035$ (women, $n_0 = 1835$) and $\hat{p} = 1755/4526 = 0.3878$ (everyone). The two-proportion statistic is

$$z = \frac{\hat{p}_1 - \hat{p}_0}{\sqrt{\hat{p}(1-\hat{p})(1/n_1 + 1/n_0)}} = \frac{0.1416}{\sqrt{0.3878 \cdot 0.6122\,(1/2691 + 1/1835)}} = \frac{0.1416}{0.01475} = 9.60,$$

where $\hat{p}_1 - \hat{p}_0 = 0.4452 - 0.3035 = 0.1416$ (from unrounded rates), $\hat{p}(1-\hat{p}) = 0.2374$, $1/2691 + 1/1835 = 0.000372 + 0.000545 = 0.000917$, and $\sqrt{0.2374 \cdot 0.000917} = \sqrt{0.000218} = 0.01475$; the two-sided p-value $2(1 - \Phi(9.60)) = 7.8 \times 10^{-22}$ says the pooled gap is no fluke. *Stratified test.* In department k write the 2×2 table as a_k = men admitted, b_k = men rejected, c_k = women admitted, d_k = women rejected, $n_k = a_k + b_k + c_k + d_k$ (e.g. department A: $a = 512$, $b = 825 - 512 = 313$, $c = 89$, $d = 108 - 89 = 19$, $n = 933$; all six departments in Table 12.5). Over the six departments 1198 men and 557 women were admitted and $2691 - 1198 = 1493$ men and $1835 - 557 = 1278$ women rejected, so the crude (pooled) odds ratio of admission, men vs. women, is $\frac{1198/1493}{557/1278} = \frac{1198 \cdot 1278}{1493 \cdot 557} = 1.841$. The Mantel–Haenszel common odds ratio combines the departmental tables,

$$\widehat{\mathrm{OR}}_{MH} = \frac{\sum_k a_k d_k / n_k}{\sum_k b_k c_k / n_k} = \frac{145.78}{161.14} = 0.905,$$

and the Cochran–Mantel–Haenszel statistic for "no sex effect in any department" compares $\sum_k a_k$ with its expectation under that hypothesis, $\mathbb{E}a_k = (a_k + b_k)(a_k + c_k)/n_k$, using $\mathrm{Var}a_k = \frac{(a_k+b_k)(c_k+d_k)(a_k+c_k)(b_k+d_k)}{n_k^2(n_k-1)}$:

$$\mathrm{CMH} = \frac{\left(|\sum_k a_k - \sum_k \mathbb{E}a_k| - \frac{1}{2}\right)^2}{\sum_k \mathrm{Var}a_k} = \frac{(|1198 - 1213.36| - 0.5)^2}{154.69} = \frac{14.857^2}{154.69} = 1.427,$$

with $p = P(\chi^2_1 > 1.427) = 0.232$. Within departments the men's odds of admission are slightly *lower* (MH odds ratio below 1), and the evidence for any common sex difference is weak. Department by department (last column of Table 12.5, the same two-proportion z computed inside each department), only Department A shows a clear difference, and it favours women ($z = -4.15$); in B–F $|z| \leq 1.00$. Both analyses are sound; they answer *different questions* (total association vs. association within department), and choosing between them is a causal decision.

PYTHON LAB · s12_examples.py · ex_sampling_noise()

```
p1 = 0.4452, p0 = 0.3035, difference = 0.1416, pooled p = 0.3878, SE =
0.01475
pooled z = 9.60 (p = 7.8e-22); crude OR = (1198*1278)/(1493*557) = 1.841
sums:  sum ad/n = 145.78, sum bc/n = 161.14, sum a = 1198, sum E[a] =
1213.36, sum Var[a] = 154.69
CMH: |1198 - 1213.36| - 0.5 = 14.857; squared = 220.74; / 154.69
Mantel-Haenszel OR = 0.905; CMH chi-square (continuity corrected) = 1.427, p
= 0.232
```

Table 12.5. Ingredients of the Mantel–Haenszel analysis of Table 12.1: a, b = men admitted/rejected, c, d = women admitted/rejected, $n = a+b+c+d$, $\mathbb{E}a = (a+b)(a+c)/n$, $\mathrm{Var}a = \frac{(a+b)(c+d)(a+c)(b+d)}{n^2(n-1)}$; z is the two-proportion statistic (men minus women) within the department. Printed by `ex_sampling_noise()`.

Dept	a	b	c	d	n	ad/n	bc/n	$\mathbb{E}a$	$\mathrm{Var}a$	z
A	512	313	89	19	933	10.43	29.86	531.43	21.91	−4.15
B	353	207	17	8	585	4.83	6.02	354.19	5.57	−0.50
C	120	205	202	391	918	51.11	45.11	114.00	47.86	+0.87
D	138	279	131	244	792	42.52	46.15	141.63	44.34	−0.55
E	53	138	94	299	584	27.14	22.21	48.08	24.25	+1.00
F	22	351	24	317	714	9.77	11.80	24.03	10.75	−0.62
Sum	1198					145.78	161.14	1213.36	154.69	

PART IV · MATHEMATICAL STATISTICS

12.10 Collider Bias Quantified

How strong is the association created by selection? For Gaussian variables and a linear selection rule it can be computed in closed form. Let ϕ, Φ be the standard normal density and cdf and $\lambda(a) = \phi(a)/(1-\Phi(a))$ the inverse Mills ratio.

Lemma 12.1 (Truncated normal moments)**.** *If $U \sim N(0,1)$ and $a \in \mathbb{R}$, then $\mathbb{E}[U \mid U > a] = \lambda(a)$ and $\mathrm{Var}(U \mid U > a) = 1 + a\lambda(a) - \lambda(a)^2 \in (0,1)$.*

The proof (integration by parts and $\lambda(a) > a$) is in Section 12.17.

Theorem 12.2 (Berkson correlation for Gaussian selection)**.** *Let $X, Y \overset{iid}{\sim} N(0,1)$, $c \in \mathbb{R}$,*

$a = c/\sqrt{2}$ *and* $\lambda = \lambda(a)$*. Then*

$$\operatorname{Corr}(X, Y \mid X + Y > c) = \frac{a\lambda - \lambda^2}{2 + a\lambda - \lambda^2} \in (-1, 0). \tag{12.4}$$

In particular the correlation is strictly negative for every threshold, tends to 0 *as* $c \to -\infty$ *(no selection), to* -1 *as* $c \to \infty$*, and equals* $-1/(\pi - 1) \approx -0.4669$ *at* $c = 0$*.*

Proof. Put $S = X+Y$ and $D = X-Y$. (S, D) is bivariate normal with $\operatorname{Cov}(S, D) = \operatorname{Var}X - \operatorname{Var}Y = 0$, hence $S \perp\!\!\!\perp D$ and the event $\{S > c\}$ leaves the law of $D \sim N(0, 2)$ unchanged. Write $X = (S + D)/2$, $Y = (S - D)/2$ and let $v = \operatorname{Var}(S \mid S > c)$. Using $\operatorname{Cov}(S, D \mid S > c) = 0$,

$$\operatorname{Cov}(X, Y \mid S > c) = \tfrac{1}{4}\big(v - 2\big), \qquad \operatorname{Var}(X \mid S > c) = \operatorname{Var}(Y \mid S > c) = \tfrac{1}{4}\big(v + 2\big),$$

so the correlation is $(v-2)/(v+2)$. Since $S/\sqrt{2} \sim N(0, 1)$ and $\{S > c\} = \{S/\sqrt{2} > a\}$, Lemma 12.1 gives $v = 2(1 + a\lambda - \lambda^2)$, and substitution yields Eq. (12.4). With $t = a\lambda - \lambda^2$ the lemma gives $t \in (-1, 0)$, so $t/(2 + t) \in (-1, 0)$. As $a \to -\infty$, $\lambda \to 0$ and $a\lambda \to 0$, so $t \to 0$; as $a \to \infty$, the truncated variance $1 + t \to 0$, so $t \to -1$ and the correlation tends to -1. At $c = 0$, $\lambda(0) = \phi(0)/\frac{1}{2} = \sqrt{2/\pi}$, so $t = -2/\pi$ and the correlation is $(-2/\pi)/(2 - 2/\pi) = -1/(\pi - 1)$. □

Worked Example 12.5 (Berkson's paradox by simulation)**.** *(Simulation:* `berkson_sim()`*,* $n = 4 \times 10^5$ *pairs* $X, Y \overset{iid}{\sim} N(0, 1)$*, seed 7.)* Keep the pairs with $X + Y > c$ and compute their sample correlation; compare with Eq. (12.4).

1. $c = 0$, so $a = 0$: theory keeps $1 - \Phi(0) = 0.5$ of the population (simulated 0.5002); $\lambda(0) = \phi(0)/0.5 = \sqrt{2/\pi} = 0.7979$, $t = a\lambda - \lambda^2 = 0 - 2/\pi = -0.6366$, and the correlation is $t/(2 + t) = -0.6366/1.3634 = -0.4669$ (simulated -0.4692).
2. $c = \sqrt{2}$, so $a = 1$: theory keeps $1 - \Phi(1) = 0.15866$ (simulated 0.1584); $\lambda(1) = \phi(1)/(1-\Phi(1)) = 0.24197/0.15866 = 1.5251$, $\lambda^2 = 2.3260$, $t = 1.5251 - 2.3260 = -0.8009$, $\operatorname{Var}(S \mid S > c) = 2(1 + t) = 0.3982$, and the correlation is $t/(2 + t) = -0.8009/1.1991 = -0.6679$ (simulated -0.6724).

Stricter selection — a more elite programme, a sicker hospital population — manufactures a *stronger* spurious trade-off between two causes that are independent in the population.

PYTHON LAB · `s12_examples.py` · `ex_berkson()`

```
c = 0.000:  kept 0.5002 of population, theory corr = -0.4669, simulated corr
= -0.4692
c = 1.414:  kept 0.1584 of population, theory corr = -0.6679, simulated corr
= -0.6724
details at c = sqrt(2):  lambda(1) = 1.5251, lambda^2 = 2.3260, t = -0.8009,
Var(S|S>c) = 0.3982
```

PART IV · MATHEMATICAL STATISTICS

12.11 Collapsibility: Confounding Is Not the Only Reason Measures Change

Definition 12.9 (Collapsibility)**.** An effect measure $\theta(p_1, p_0)$ (risk difference $p_1 - p_0$, risk ratio p_1/p_0, odds ratio $\frac{p_1/(1-p_1)}{p_0/(1-p_0)}$) is *collapsible over* Z if, whenever $Z \perp\!\!\!\perp X$ and the stratum measures are all equal to a common value θ^*, the marginal measure also equals θ^*.

Proposition 12.2 (RD is collapsible, OR is not)**.** *If $Z \perp\!\!\!\perp X$, the marginal risk difference is the $P(z)$-weighted average of the stratum risk differences, so the RD is collapsible; the RR is collapsible as well. The odds ratio is not collapsible: there are distributions with $Z \perp\!\!\!\perp X$, identical stratum odds ratios and a different marginal odds ratio.*

Proof. With $Z \perp\!\!\!\perp X$, $p_x = \sum_z P(z)p_{xz}$, so $p_1 - p_0 = \sum_z P(z)(p_{1z} - p_{0z})$, which equals θ^* if all strata do. If every stratum has $p_{1z}/p_{0z} = \theta^*$ (with $p_{0z} > 0$), then $p_1 = \sum_z P(z)\,\theta^* p_{0z} = \theta^* p_0$, so the marginal RR is θ^* too. For the odds ratio, the counterexample of Worked Example 12.6 suffices. The underlying reason is that $p \mapsto \log \frac{p}{1-p}$ is non-linear: the logit of an average is not the average of logits. □

Worked Example 12.6 (Non-collapsibility of the odds ratio without confounding)**.** (Illustrative.) Let $Z \sim \text{Bern}(1/2)$ independent of X, with $P(Y = 1 \mid X, Z)$ equal to 0.8 vs. 0.5 (treated vs. untreated) when $Z = 0$ and 0.5 vs. 0.2 when $Z = 1$. Stratum $Z = 0$: $\text{OR} = \frac{0.8/0.2}{0.5/0.5} = \frac{4}{1} = 4$, $\text{RD} = 0.8 - 0.5 = 0.3$; stratum $Z = 1$: $\text{OR} = \frac{0.5/0.5}{0.2/0.8} = \frac{1}{0.25} = 4$, $\text{RD} = 0.5 - 0.2 = 0.3$. Marginally (using $Z \perp\!\!\!\perp X$) $p_1 = 0.5(0.8) + 0.5(0.5) = 0.65$ and $p_0 = 0.5(0.5) + 0.5(0.2) = 0.35$, so $\text{RD} = 0.65 - 0.35 = 0.3$ (collapsible), but the marginal odds are $0.65/0.35 = 1.8571$ and $0.35/0.65 = 0.5385$, giving $\text{OR} = 1.8571/0.5385 = 3.449 < 4$. There is *no confounding* here ($Z \perp\!\!\!\perp X$, so $P(y \mid x) = P(y \mid \text{do}(x))$); the change in the odds ratio is a property of the measure. Practical consequence: in logistic regression, adding a prognostic covariate that is independent of a randomized treatment changes the treatment coefficient ($\log 4 = 1.3863$ vs. marginal 1.2381) without signalling any bias.

PYTHON LAB · s12_examples.py · ex_noncollapsible()

```
strata:  OR=4.000, RD=0.300; marginal:  P(Y|X=1)=0.650, P(Y|X=0)=0.350,
OR=3.449, RD=0.300
marginal odds:  0.65/0.35 = 1.8571, 0.35/0.65 = 0.5385; ratio = 3.449
conditional log-OR = log 4 = 1.3863; marginal log-OR = 1.2381
```

PART IV · MATHEMATICAL STATISTICS

12.12 Omitted Variables, Proxies and M-Bias in Linear Models

Proposition 12.3 (Omitted-variable bias). *Suppose* $Y = \alpha + \beta X + \gamma Z + \varepsilon$ *with* $\mathbb{E}[\varepsilon \mid X, Z] = 0$ *and* $\mathrm{Var}X > 0$. *The population slope of the* short *regression of* Y *on* X *alone is*

$$b_{Y \cdot X} = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}X} = \beta + \gamma\,\delta, \qquad \delta = \frac{\mathrm{Cov}(X, Z)}{\mathrm{Var}X}, \tag{12.5}$$

and the OLS estimator $\hat{b}_{Y \cdot X}$ *from an i.i.d. sample converges in probability to* $\beta + \gamma\delta$.

Proof. $\mathrm{Cov}(X, Y) = \beta \mathrm{Var}X + \gamma \mathrm{Cov}(X, Z) + \mathrm{Cov}(X, \varepsilon)$ and $\mathrm{Cov}(X, \varepsilon) = \mathbb{E}[X\varepsilon] - \mathbb{E}X\,\mathbb{E}\varepsilon = \mathbb{E}[X\,\mathbb{E}(\varepsilon \mid X, Z)] = 0$. Divide by $\mathrm{Var}X$. For the estimator, $\hat{b} = \widehat{\mathrm{Cov}}(X, Y)/\widehat{\mathrm{Var}}(X)$; the sample moments converge in probability to their population counterparts by the weak law of large numbers, and the continuous mapping theorem gives $\hat{b} \xrightarrow{p} \beta + \gamma\delta$. □

MATHEMATICAL STATISTICS · Reading the omitted-variable bias formula

The bias $\gamma\delta$ is the product of the effect of the omitted variable on the outcome (γ) and its regression on the included regressor (δ) — exactly the two arrows of the back-door path $X \leftarrow Z \rightarrow Y$, the linear analogue of the composition term of Theorem 12.1. Its sign is $\mathrm{sign}(\gamma)\,\mathrm{sign}(\delta)$: omitting ability ($\gamma > 0$) that correlates positively with schooling ($\delta > 0$) overstates the return to schooling. If $Z \perp\!\!\!\perp X$ ($\delta = 0$) the short regression is unbiased for β, which is the linear-model content of Theorem 12.3. Wright's path-tracing rules (Session 13) generalise Eq. (12.5) to arbitrary linear DAGs.

Worked Example 12.7 (Omitted-variable bias by simulation). (Illustrative.) $Z \sim N(0, 1)$, $X = 0.8Z + U$, $Y = X + 2Z + \varepsilon$ with U, ε independent standard normal noises, so $\beta = 1$ and $\gamma = 2$. Then $\mathrm{Var}X = 0.8^2\mathrm{Var}Z + \mathrm{Var}U = 0.64 + 1 = 1.64$ and $\mathrm{Cov}(X, Z) = 0.8\mathrm{Var}Z = 0.8$, so $\delta = 0.8/1.64$ and Eq. (12.5) predicts a short slope of $1 + 2(0.8)/1.64 = 1 + 1.6/1.64 = 1 + 0.9756 = 1.9756$, almost double $\beta = 1$. *Simulation* ($n = 10^5$ draws, seed 7): the short regression of Y on X gives slope 1.9721, the long regression of Y on (X, Z) gives 0.9944.

PYTHON LAB · s12_examples.py · ex_ovb()

```
simulation:  n = 100000, seed 7
gamma*Cov/Var = 2.0*0.8/1.64 = 1.6/1.64 = 0.9756; theory plim 1.9756
simulated short-regression slope = 1.9721; long regression (with Z) = 0.9944
```

Worked Example 12.8 (Residual confounding with a noisy proxy). *(Analytical and numerical, illustrative.)* Let $Z \sim N(0, 1)$, $X = Z + U$, $Y = X + 2Z + \varepsilon$ (U, ε independent standard normal; true effect of X is 1), and suppose we only observe a proxy $W = Z + e$ with independent noise $e \sim N(0, s^2)$. Its *reliability* is $\mathrm{Var}Z/\mathrm{Var}W = 1/(1 + s^2)$.

1. *Reduction.* Because $Y = X + 2Z + \varepsilon$ and ε is uncorrelated with (X, W), the population regression of Y on (X, W) has X-coefficient $1 + 2\pi_X$, where π_X is the X-coefficient in the regression of Z on (X, W).
2. *Moments.* $\mathrm{Var}X = 1 + 1 = 2$, $\mathrm{Var}W = 1 + s^2$, $\mathrm{Cov}(X, W) = \mathrm{Cov}(X, Z) = \mathrm{Cov}(W, Z) =$

$\mathrm{Var}Z = 1$.

3. *Normal equations.* For two regressors,

$$\pi_X = \frac{\mathrm{Var}W\,\mathrm{Cov}(X,Z) - \mathrm{Cov}(X,W)\mathrm{Cov}(W,Z)}{\mathrm{Var}X\,\mathrm{Var}W - \mathrm{Cov}(X,W)^2} = \frac{(1+s^2)\cdot 1 - 1\cdot 1}{2(1+s^2)-1} = \frac{s^2}{1+2s^2},$$

so the bias is $2\pi_X = 2s^2/(1+2s^2)$.

4. *Numbers.* $s^2 = 0$ (perfect proxy): bias 0. Reliability 0.8 ($s^2 = 0.25$): $0.5/1.5 = 0.3333$. Reliability 0.5 ($s^2 = 1$): $2/3 = 0.6667$. Reliability 0.2 ($s^2 = 4$): $8/9 = 0.8889$. No adjustment at all: slope $\mathrm{Cov}(X,Y)/\mathrm{Var}X = (\mathrm{Var}X + 2\mathrm{Cov}(X,Z))/\mathrm{Var}X = (2+2)/2 = 2$, bias 1.

PYTHON LAB · s12_examples.py · ex_proxy()

```
Var(e) = 0.00, 0.25, 1.00, 4.00:  coef.  on X = 1.0000, 1.3333, 1.6667,
1.8889; no adjustment:  2.0000
```

Worked Example 12.9 (M-bias: adjusting for a pre-treatment collider)**.** (Illustrative.) Let U_1, U_2 be independent $N(0,1)$, $X = U_1 + e_X$, $M = U_1 + U_2 + e_M$, $Y = X + U_2 + e_Y$, with e_X, e_M, e_Y independent $N(0,1)$ noises (true effect of X on Y is 1). There is no open back-door path ($X \leftarrow U_1 \rightarrow M \leftarrow U_2 \rightarrow Y$ is blocked at the collider M), so the regression of Y on X alone is unbiased: $\mathrm{Var}X = \mathrm{Var}U_1 + \mathrm{Var}e_X = 2$, $\mathrm{Cov}(X,Y) = \mathrm{Cov}(X, X + U_2 + e_Y) = \mathrm{Var}X = 2$, slope $2/2 = 1$. Adjusting for M: $\mathrm{Var}M = 1+1+1 = 3$, $\mathrm{Cov}(X,M) = \mathrm{Var}U_1 = 1$, $\mathrm{Cov}(M,Y) = \mathrm{Cov}(M,X) + \mathrm{Cov}(M,U_2) = 1+1 = 2$, so the coefficient of X is

$$\frac{\mathrm{Var}M\,\mathrm{Cov}(X,Y) - \mathrm{Cov}(X,M)\mathrm{Cov}(M,Y)}{\mathrm{Var}X\mathrm{Var}M - \mathrm{Cov}(X,M)^2} = \frac{3\cdot 2 - 1\cdot 2}{2\cdot 3 - 1} = \frac{4}{5} = 0.8.$$

Conditioning on M opened $X \leftarrow U_1 \rightarrow M \leftarrow U_2 \rightarrow Y$ and introduced a bias of -0.2.

PYTHON LAB · s12_examples.py · ex_mbias()

```
population moments:  Var X = 2, Var M = 3, Cov(X,Y) = 2, Cov(X,M) = 1,
Cov(M,Y) = 2
Y on X: 1.0000; Y on X and M: coefficient on X = 0.8000 (bias -0.2000)
```

PART IV · MATHEMATICAL STATISTICS

12.13 Why Randomization Works

Potential outcomes (Session 15) make the argument transparent. Each unit has two potential outcomes $Y(1), Y(0)$; we observe $Y = Y(T)$ (*consistency*). The average treatment effect is $\mathrm{ATE} = \mathbb{E}[Y(1) - Y(0)]$ and the effect on the treated is $\mathrm{ATT} = \mathbb{E}[Y(1) - Y(0) \mid T = 1]$.

Theorem 12.3 (Randomization identifies the ATE)**.** *For any assignment mechanism with* $0 < P(T=1) < 1$,

$$\mathbb{E}[Y \mid T=1] - \mathbb{E}[Y \mid T=0] = \mathrm{ATT} + \underbrace{\big(\mathbb{E}[Y(0) \mid T=1] - \mathbb{E}[Y(0) \mid T=0]\big)}_{\textit{selection bias}}. \tag{12.6}$$

If T is assigned by an independent randomization device, then $T \perp\!\!\!\perp (Y(0), Y(1))$, the selection-bias term is zero, $\mathrm{ATT} = \mathrm{ATE}$*, and the difference in means equals the ATE. Moreover, with n_1 treated and n_0 control units drawn i.i.d. from the arms, the difference of sample means $\hat{\tau}$ is unbiased with* $\mathrm{Var}(\hat{\tau}) = \sigma_1^2/n_1 + \sigma_0^2/n_0$*, where* $\sigma_t^2 = \mathrm{Var}(Y(t))$.

Proof. By consistency, $\mathbb{E}[Y \mid T = t] = \mathbb{E}[Y(t) \mid T = t]$. Adding and subtracting $\mathbb{E}[Y(0) \mid T = 1]$, $\mathbb{E}[Y(1) \mid T = 1] - \mathbb{E}[Y(0) \mid T = 0] = \big(\mathbb{E}[Y(1) \mid T = 1] - \mathbb{E}[Y(0) \mid T = 1]\big) + \big(\mathbb{E}[Y(0) \mid T = 1] - \mathbb{E}[Y(0) \mid T = 0]\big)$, which is Eq. (12.6). If T is produced by a device independent of everything else (e.g. a coin, $T = \mathbb{1}\{R < q\}$ with $R \perp\!\!\!\perp (Y(0), Y(1))$), then T is a measurable function of R and hence $T \perp\!\!\!\perp (Y(0), Y(1))$. Then $\mathbb{E}[Y(t) \mid T = t'] = \mathbb{E}[Y(t)]$ for all t, t', so the bias term is 0 and $\mathrm{ATT} = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] = \mathrm{ATE}$. For the variance: the two sample means are independent, unbiased for $\mathbb{E}[Y(1)]$ and $\mathbb{E}[Y(0)]$, with variances σ_1^2/n_1 and σ_0^2/n_0 because, given $T = t$, the observed Y has the law of $Y(t)$. □

Worked Example 12.10 (Observational vs. randomized assignment)**.** (Illustrative.) $Z \sim \mathrm{Bern}(0.5)$ (say, severity), $Y = 1 + 2T + 3Z + \varepsilon$ with $\varepsilon \sim N(0, 1)$ independent of (T, Z), so $Y(1) - Y(0) = 2$ for every unit and $\mathrm{ATE} = \mathrm{ATT} = 2$.

1. *Observational assignment.* The sicker are treated more: $P(T = 1 \mid Z = 1) = 0.8$, $P(T = 1 \mid Z = 0) = 0.2$, so $P(T = 1) = 0.5(0.8) + 0.5(0.2) = 0.5$. By Bayes, $\mathbb{E}[Z \mid T = 1] = P(Z = 1 \mid T = 1) = 0.5(0.8)/0.5 = 0.8$ and $\mathbb{E}[Z \mid T = 0] = 0.5(0.2)/0.5 = 0.2$.
2. *Selection bias.* $Y(0) = 1 + 3Z + \varepsilon$, so the bias term of Eq. (12.6) is $\mathbb{E}[Y(0) \mid T = 1] - \mathbb{E}[Y(0) \mid T = 0] = 3(0.8 - 0.2) = 1.8$, and the difference in means tends to $\mathrm{ATT} + 1.8 = 2 + 1.8 = 3.8$.
3. *Randomized assignment.* With $P(T = 1) = 0.5$ independently of Z, the bias term is 0. Within an arm $\mathrm{Var}(Y \mid T) = \mathrm{Var}(3Z + \varepsilon) = 9(0.5 \cdot 0.5) + 1 = 3.25$; with $n = 400$ units the arms have $n_t \approx 200$, and Theorem 12.3 gives $\mathrm{sd}(\hat{\tau}) \approx \sqrt{3.25/200 + 3.25/200} = 0.180$.
4. *Simulation* (`randomization_sims()`: 2000 replications of $n = 400$ units, seed 7; "sd" is the standard deviation of the 2000 estimates). Observational difference in means: average 3.803 (sd 0.157); randomized: average 2.004 (sd 0.186, close to the predicted 0.180).

PYTHON LAB · `s12_examples.py` · `ex_randomization()`

```
Bayes:  E[Z|T=1] = 0.8, E[Z|T=0] = 0.2 -> difference 0.6, selection bias
3*0.6 = 1.8
simulation:  2000 replications, n = 400 units each, seed 7
observational mean 3.803 (sd 0.157); RCT mean 2.004 (sd 0.186); RCT theory
sd 0.180
```

Randomization removes confounding in expectation, not selection: drop-out after randomization re-creates a collider $T \to S \leftarrow Y$.

PART IV · MATHEMATICAL STATISTICS

12.14 Worked Examples

Worked Example 12.11 (Reconstructing the Berkeley reversal arithmetically)**.** Using only Department A (825 men, 512 admitted; 108 women, 89 admitted) and Department C (325 men, 120 admitted; 593 women, 202 admitted), verify that women's aggregate admission rate can fall below men's even though women's rate is higher in Department A specifically.
Solution. In Department A alone: men $512/825 = 62.1\%$, women $89/108 = 82.4\%$ — women higher. In Department C alone: men $120/325 = 36.9\%$, women $202/593 = 34.1\%$ — men slightly higher, but the more important asymmetry is where the applicants concentrate: department C received far more of its total applicant pool from women (593) than men (325), and C is intrinsically a lower-admission-rate department for everyone. Aggregating A and C together: men $(512+120)/(825+325) = 632/1150 = 55.0\%$, women $(89+202)/(108+593) = 291/701 = 41.5\%$ — the aggregate now favors men even though women did much better in the higher-admitting department, purely because women's applications concentrated more heavily in the lower-admitting one. The weighted-average form makes this explicit: $593/701 = 84.6\%$ of the women but only $325/1150 = 28.3\%$ of the men applied to C, so

$$\begin{aligned} \text{women: } & 0.154(0.824) + 0.846(0.341) = 0.415, \\ \text{men: } & 0.717(0.621) + 0.283(0.369) = 0.550, \end{aligned}$$

where $0.154 = 108/701$ and $0.717 = 825/1150$ are the shares applying to A. This is Simpson's paradox reduced to two departments and pure arithmetic: no discrimination coefficient is needed to produce the gap, only a differential rate of applying to harder departments.

PYTHON LAB · s12_examples.py · ex_berkeley_two_depts()

```
A+C aggregate:  men 632/1150 = 0.550, women 291/701 = 0.415
share applying to C: men 325/1150 = 0.283, women 593/701 = 0.846
weighted form, women:  0.154*0.824 + 0.846*0.341 = 0.415
weighted form, men:  0.717*0.621 + 0.283*0.369 = 0.550
```

Worked Example 12.12 (A marketing analogue)**.** An online retailer notices that customers who were shown a promotional ad convert (make a purchase) 19% of the time, versus 9% for those not shown the ad — seemingly a strong case for showing the ad to everyone. But the ad-targeting algorithm preferentially shows ads to customers with high prior purchase intent (say, based on recent browsing activity), and high-intent customers convert at a high rate *regardless* of the ad. If, hypothetically, high-intent customers convert at 25% whether or not they see the ad, and low-intent customers convert at 5% whether or not they see the ad, but 70% of ad-shown customers are high-intent versus only 20% of not-shown customers, then the naive comparison recovers exactly the "effect" quoted above ($0.7(0.25) + 0.3(0.05) = 0.19$ vs. $0.2(0.25) + 0.8(0.05) = 0.09$) with *zero true causal effect of the ad at all* — purchase intent is a confounder (it drives both ad exposure and purchase), exactly as stone size was for kidney stones. (Berkeley has the same arithmetic, but there department is a mediator, not a confounder; Section 12.2.) In the language of Theorem 12.1 the within-stratum effect is 0 (in each intent stratum the two conversion rates coincide) and the whole gap is composition: with $Z = 1$ for high intent, $(w_1^1 - w_1^0)(p_{01} - p_{00}) = (0.7 - 0.2)(0.25 - 0.05) = 0.5 \cdot 0.20 = 0.10$,

i.e. all 10 points of $0.19 - 0.09$. This is the vignette we will resolve properly with a formal identification argument in Sessions 14 and 15.

PYTHON LAB · s12_examples.py · ex_marketing()

```
P(convert | ad) = 0.190, P(convert | no ad) = 0.090, naive gap = 0.100
composition term (0.7 - 0.2)*(0.25 - 0.05) = 0.100; within-stratum term = 0
```

Worked Example 12.13 (Distinguishing confounding from mere sampling noise)**.** A skeptic argues both reversals above might just be small-sample flukes. Check this for Berkeley.
Solution. The pooled gap is real: with 2691 men and 1835 women in the six departments (8442 and 4321 campus-wide), the 14.2-point gap $0.4452 - 0.3035 = 0.1416$ gives $z = 9.60$ (Section 12.9), far beyond any chance fluctuation. The within-department differences are a different matter: they are small and mostly within noise (Table 12.5: $|z| \leq 1.00$ in departments B–F), the one clear exception being Department A, where *women* do better ($z = -4.15$); combined, the Mantel–Haenszel odds ratio is 0.905 with CMH $p = 0.232$. So the reversal is not a fluke in either direction: a large, statistically solid pooled gap coexists with small within-department differences. The point of stratification is not to *explain away* a real effect as noise, but to reveal that the aggregated number was measuring a mixture of two different things (department-level admission difficulty and sex) that happened to move together in the data — a structural fact about how the applicant pool was formed, not a statistical fluke.

Additional Worked Examples (Instructor Supplement)

Worked Example 12.14 (A third real reversal: batting averages)**.** Simpson's paradox is not confined to admissions or medicine. In the 1995 and 1996 Major League Baseball seasons, David Justice and Derek Jeter's batting averages (hits/at-bats) were:

$$\text{1995: Justice } \tfrac{104}{411} \approx .253,\ \text{Jeter } \tfrac{12}{48} = .250; \qquad \text{1996: Justice } \tfrac{45}{140} \approx .321,\ \text{Jeter } \tfrac{183}{582} \approx .314.$$

Justice has the higher average in *both* individual seasons. Yet combined over the two years,

$$\text{Justice: } \frac{104+45}{411+140} = \frac{149}{551} \approx \mathbf{.270}, \qquad \text{Jeter: } \frac{12+183}{48+582} = \frac{195}{630} \approx \mathbf{.310},$$

and Jeter's combined average is *higher*. The arithmetic is identical to Berkeley and the kidney stones. Graphically, the season plays the role of stone size (Fig. 12.4): for a single at-bat, the season comes first and influences both *who* bats (Jeter, a rookie in 1995, had only 48 at-bats that year) and the chance of a hit (both players hit better in 1996). Jeter had $582/630 = 92.4\%$ of his at-bats in 1996, against $140/551 = 25.4\%$ for Justice, so his combined average is weighted toward the better year:

$$\begin{aligned} \text{Justice: } & 0.746(0.253) + 0.254(0.321) = 0.270, \\ \text{Jeter: } & 0.076(0.250) + 0.924(0.314) = 0.310 \end{aligned}$$

(Fig. 12.9, right). No baseball skill differential is needed to produce this reversal — only a differential in *how many at-bats occurred in which year*, exactly mirroring the differential application rates to departments (Berkeley) and differential treatment assignment by stone size (kidney stones).

PYTHON LAB · s12_examples.py · ex_batting()

```
1995:  Justice 104/411 = 0.253, Jeter 12/48 = 0.250
1996:  Justice 45/140 = 0.321, Jeter 183/582 = 0.314
Justice 149/551 = 0.270; weighted:  0.746*0.253 + 0.254*0.321 = 0.270
Jeter 195/630 = 0.310; weighted:  0.076*0.250 + 0.924*0.314 = 0.310
```

Worked Example 12.15 (Simpson's paradox in a continuous outcome, not just a rate)**.** (Hypothetical, for illustration.) Two companies' employees report job-satisfaction scores by seniority. Company X: 90 senior staff, average satisfaction 9.0; 10 junior staff, average satisfaction 5.0. Company Y: 10 senior staff, average satisfaction 9.5; 90 junior staff, average satisfaction 5.5. Within *each* seniority level, Company Y does better (junior: $5.5 > 5.0$; senior: $9.5 > 9.0$). Yet aggregated:

$$\text{X: } \frac{90(9.0)+10(5.0)}{100} = \frac{810+50}{100} = \mathbf{8.60}, \qquad \text{Y: } \frac{10(9.5)+90(5.5)}{100} = \frac{95+495}{100} = \mathbf{5.90}.$$

Company X's aggregate score (8.60) is far *higher* than Company Y's (5.90), reversing Y's within-group advantage in both strata. This confirms Simpson's paradox is a property of any weighted average, not merely proportions (Theorem 12.1 holds verbatim with conditional means in place of p_{xz}): Company X simply has a much more senior (and hence, within-level, happier) workforce composition, letting a systematically *worse* performer at every seniority level post the better headline number — exactly the same weighted-average arithmetic behind this session's admissions and treatment examples, here applied to a continuous score rather than a binary success rate.

PYTHON LAB · s12_examples.py · ex_continuous()

```
Company X aggregate = 8.60, Company Y aggregate = 5.90
```

PART V

From Observational Data to Acting Agents

Agentic AI, reinforcement learning, multi-agent systems and deep learning

PART V · AI CONNECTIONS

12.15 Why an AI Agent Must Climb the Ladder

An agent (course session 1) perceives, decides and *acts*; its action is an intervention, a rung-2 object. Whenever an AI system learns about the consequences of actions from data generated by someone else's policy, confounding, selection and non-collapsibility are all in play (Fig. 12.10).

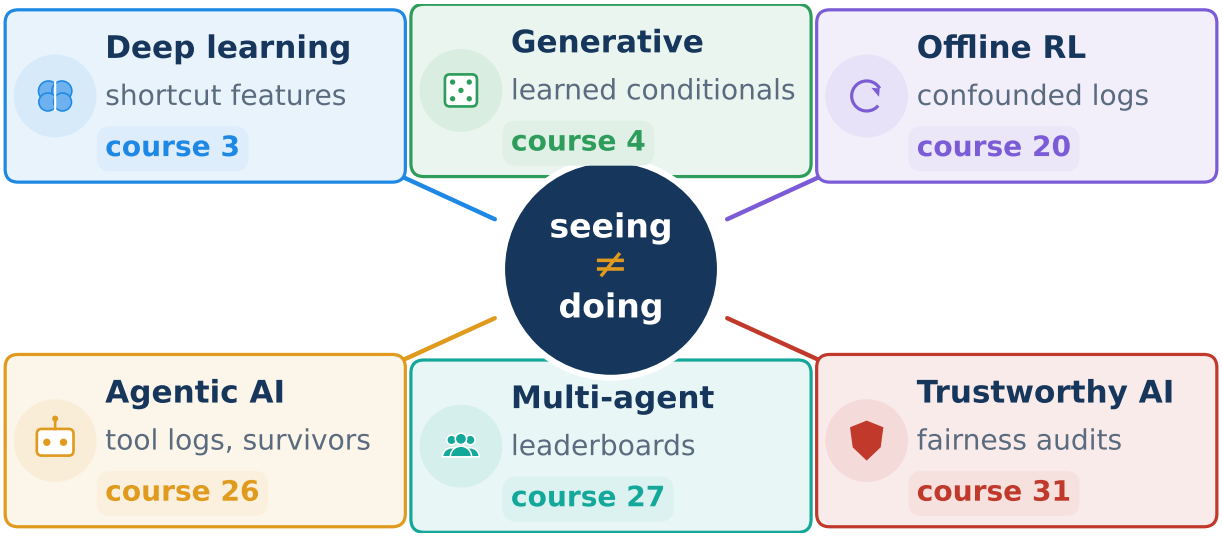


Figure 12.10. Where the seeing/doing gap reappears in the Advanced AI course.

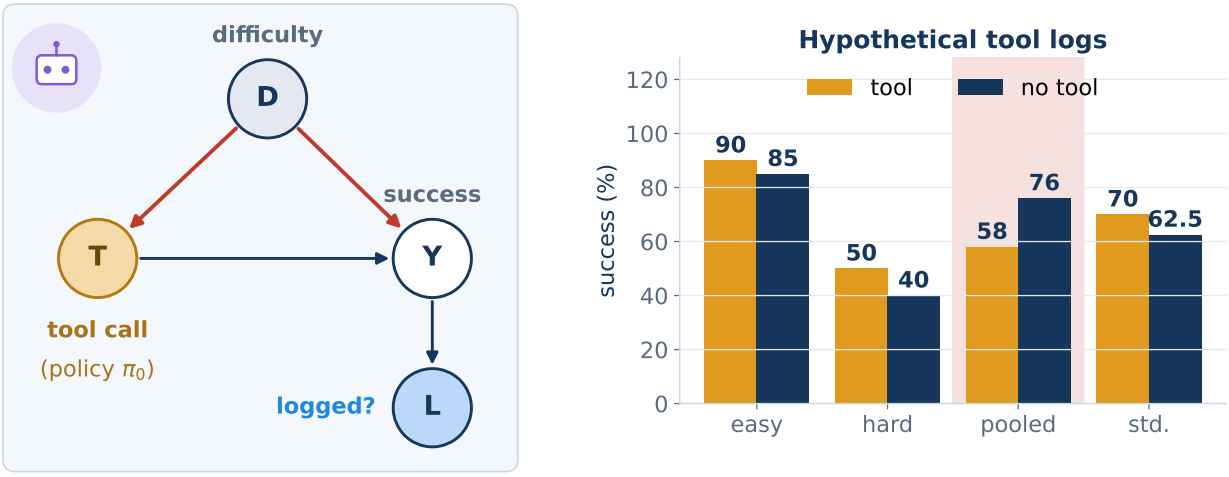


Figure 12.11. Left: the causal graph of an agent's logs. Query difficulty D confounds the old policy's tool choice T and success Y; whether a run is logged (L) may depend on Y (survivorship). Right: the hypothetical logs of Worked Example 12.16.

12.15.1 Agentic AI: learning from logged tool use

Worked Example 12.16 (Does the retrieval tool help? (hypothetical logs))**.** An LLM agent's past policy π_0 called a retrieval tool mostly on hard queries. Logs: easy queries — with tool 90/100, without 340/400; hard queries — with tool 200/400, without 40/100. Pooled success is $(90+200)/(100+400) = 290/500 = 0.580$ with the tool and $(340+40)/(400+100) = 380/500 = 0.760$ without: a naive analysis, or a fine-tuning pipeline that imitates high-reward trajectories, would learn "don't call the tool". Within each difficulty level the tool helps: easy $90/100 = 0.90$ vs. $340/400 = 0.85$ (+5 points), hard $200/400 = 0.50$ vs. $40/100 = 0.40$ (+10 points). The log contains $100+400 = 500$ easy and 500 hard queries, so $P(\text{easy}) = 500/1000 = 0.5$; standardizing to this mix gives tool $0.5(0.90) + 0.5(0.50) = 0.700$ vs. no tool $0.5(0.85) + 0.5(0.40) = 0.625$, an estimated effect of $0.700 - 0.625 = +0.075$ — valid only if difficulty is the *only* confounder and is recorded.

PYTHON LAB · s12_examples.py · ex_agent_logs()

```
aggregate tool:  290/500 = 0.580; aggregate no_tool:  380/500 = 0.760;
P(easy) = 500/1000 = 0.50
standardised (P(easy) = 0.50):  tool 0.700, no tool 0.625, effect +0.075
```

Worked Example 12.17 (Survivorship in agent evaluation (hypothetical))**.** An agent truly solves 40% of tasks; every success is logged but only a fraction $q = 0.25$ of failures (the rest crash silently). Per attempted task, the log receives a success with probability $p = 0.40$ and a failure with probability $(1-p)q = 0.60(0.25) = 0.15$, so the logged success rate is $\frac{p}{p+(1-p)q} = \frac{0.40}{0.40+0.15} = \frac{0.40}{0.55} = 0.727$, and a version with *better* telemetry (larger q) will look *worse*.

PYTHON LAB · s12_examples.py · ex_survivorship()

```
true success 0.40; failures logged with prob 0.25:  logged success rate =
0.727
= 0.40/(0.40 + 0.60*0.25) = 0.40/(0.40 + 0.15) = 0.40/0.55
```

LINK TO COURSE SESSION 26 · Agentic AI

Tool use, memory and reflection in LLM agents are increasingly tuned on the agents' own logs: observational data from an earlier policy, confounded by task difficulty and often filtered by success before storage. "Does calling T help?" is a rung-2 question; Sessions 14–15 give the tools (adjustment, IPW, A/B tests of agent versions).

12.15.2 Reinforcement learning: confounded offline data

In a Markov decision process (MDP) the value of an action is interventional: $Q^\pi(s,a) = \mathbb{E}[G_t \mid S_t = s, \mathrm{do}(A_t = a)]$, where G_t is the return (the discounted sum of the rewards after time t) and all actions after A_t follow the policy π (course session 17). Online, the agent chooses a itself, so the action is randomized by the agent's own policy and $P(\cdot \mid a) = P(\cdot \mid \mathrm{do}(a))$. Offline RL (course session 20) learns from logs produced by a behaviour policy that may have seen information the learner does not.

Worked Example 12.18 (A confounded logged bandit (hypothetical))**.** A context is good or bad with probability 1/2; the human operator who generated the logs saw it and chose arm 1 with probability 0.9 in good contexts and 0.1 in bad ones. Mean rewards: arm 1: 0.7 (good), 0.2 (bad); arm 0: 0.8 (good), 0.3 (bad) — arm 0 is better in *every* context. The learner does not observe the context.

1. *Who is in each arm.* $P(a=1) = 0.5(0.9) + 0.5(0.1) = 0.5$; by Bayes $P(\text{good} \mid a=1) = 0.5(0.9)/0.5 = 0.9$ and $P(\text{good} \mid a=0) = 0.5(0.1)/0.5 = 0.1$.
2. *Naive logged means.* Arm 1: $0.9(0.7) + 0.1(0.2) = 0.63 + 0.02 = 0.65$; arm 0: $0.1(0.8) + 0.9(0.3) = 0.08 + 0.27 = 0.35$.
3. *Interventional values* (context keeps $P(\text{good}) = 0.5$): $\mathbb{E}[R \mid \text{do}(1)] = 0.5(0.7) + 0.5(0.2) = 0.45$, $\mathbb{E}[R \mid \text{do}(0)] = 0.5(0.8) + 0.5(0.3) = 0.55$.

Greedy offline policy improvement picks the worse arm, and because the confounder is unobserved no reweighting of these logs can fix it.

PYTHON LAB · s12_examples.py · ex_offpolicy()

```
P(a=1) = 0.5*0.9 + 0.5*0.1 = 0.50; P(good | a=1) = 0.90, P(good | a=0) =
0.10
naive logged means:  arm1 0.650, arm0 0.350 (arm1 = 0.63 + 0.02, arm0 = 0.08
+ 0.27)
E[R | do(a)]:  arm1 0.450, arm0 0.550 (arm1 = 0.35 + 0.10, arm0 = 0.40 +
0.15)
```

LINK TO COURSE SESSION 20 · Deep and Model-based RL

Offline RL and off-policy evaluation assume the behaviour policy used only recorded state; importance sampling then corrects for *which* actions were taken. If a human or legacy system used unrecorded information, the logs are confounded as in Worked Example 12.18. Session 15 shows that importance-sampling OPE is inverse propensity weighting.

12.15.3 Multi-agent AI: Simpson's paradox on leaderboards

Worked Example 12.19 (A leaderboard reversal (hypothetical))**.** In an arena where stronger-looking agents are matched with harder tasks, agent A solves easy $45/50 = 0.90$ and hard $135/450 = 0.30$; agent B easy $340/400 = 0.85$ and hard $25/100 = 0.25$. A wins both categories, yet the pooled leaderboard reads $(45+135)/(50+450) = 180/500 = 0.360$ for A and $(340+25)/(400+100) = 365/500 = 0.730$ for B, because $450/500 = 90\%$ of A's tasks were hard against $100/500 = 20\%$ of B's. The macro (equal-weight) score — standardization to a fixed task mix with $P(\text{easy}) = P(\text{hard}) = 0.5$ — restores the order: A $0.5(0.90) + 0.5(0.30) = 0.600$ vs. B $0.5(0.85) + 0.5(0.25) = 0.550$. Matchmaking, opponent choice and task assignment all play the role of Z.

PYTHON LAB · s12_examples.py · ex_leaderboard()

```
agent A: overall 180/500 = 0.360; equal-weight (macro) score = 0.600
agent B: overall 365/500 = 0.730; equal-weight (macro) score = 0.550
```

LINK TO COURSE SESSION 27 · Multi-Agent AI

Comparing agents or protocols across a population is observational unless pairing is randomized; interacting agents also violate the "no interference" assumption of Session 15, so experiments must be designed at the group level.

12.15.4 Deep learning: shortcut learning is rung-1 learning

Worked Example 12.20 (A spurious feature beats the causal one (synthetic)). *Setup* (`ex_shortcut()`, seed 7). Labels $y \in \{0, 1\}$ are drawn with probability 1/2 each; write $\pm 1 = 2y - 1$. The label causes a noisy "core" feature $(\pm 1 + N(0, 1))$; in training, a nearly noiseless "background" feature $(\pm 1 + N(0, 0.3^2))$ agrees in sign with the label in 95% of the examples (and has the opposite sign otherwise). A logistic regression (scikit-learn default, L2 penalty with $C = 1$) is fitted on $n = 5000$ training examples and evaluated on three test sets of 5000 examples each. It weights the background more (2.85) than the core (1.95). Test accuracy is 0.958 i.i.d., 0.660 when the background agrees only 50% of the time and 0.372 when the agreement is reversed (5%); a core-only model keeps 0.844 under the 50% shift. The model learned $P(y \mid x)$, as Sessions 5–11 asked; the background is linked to the label by data collection, not by the label-generating mechanism.

PYTHON LAB · s12_examples.py · ex_shortcut()

```
setup:  n = 5000 training and 5000 per test set, seed 7, sklearn
LogisticRegression (default L2, C=1)
weights (core, spurious) = [1.95 2.85]; accuracy 0.958 / 0.660 / 0.372;
core-only 0.844
```

LINK TO COURSE SESSION 3 · Advanced Learning

Generalization under distribution shift, domain adaptation and invariant representation learning are attempts to learn features whose relation to the label is *causal* and therefore stable across environments (Peters, Janzing & Schölkopf 2017). Shortcut learning is what empirical risk minimization does when the training distribution contains a non-causal path. Likewise a conditional generative model (course session 4) samples from a learned $P(x \mid c)$; editing a given x to show "what x would look like had c been different" is a rung-3 query.

LINK TO COURSE SESSION 31 · Frontiers: Trustworthy AI

Berkeley is the founding example of fairness auditing: whether the pooled or the within-department gap measures "discrimination" depends on which paths from sex to admission are deemed legitimate.

PART V · AI CONNECTIONS

12.16 Summary

Three questions for discussion, then the formulas and pitfalls to keep.

Table 12.6. This session in the Advanced AI course.

Concept today	Where it reappears	Course session
Seeing vs. doing	action values $Q(s,a)$ are interventional; online vs. offline RL	17, 18, 20
Confounding	logged tool use of agents; behaviour policies in offline RL	20, 26
Selection / survivorship	filtered trajectories, benchmark drop-outs, memory curation	26, 31
Simpson's paradox	leaderboards, populations of agents, matchmaking	27
Causal graph	SCMs, do-calculus, effect estimation	13, 14, 15
Randomization	A/B tests of agent versions; exploration as randomization	18, 26

Table 12.7. Cheat sheet of key formulas.

Object	Formula
Observational vs. interventional	$P(y \mid x) = \sum_z P(y \mid x,z)P(z \mid x)$ vs. $P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid x,z)P(z)$
Simpson decomposition	$p_1 - p_0 = \sum_z w_z^1(p_{1z} - p_{0z}) + \sum_z (w_z^1 - w_z^0)p_{0z}$
No reversal if	$Z \perp\!\!\!\perp X$: $p_1 - p_0 = \sum_z P(z)(p_{1z} - p_{0z})$
Berkson correlation	$\mathrm{Corr}(X,Y \mid X+Y>c) = \frac{a\lambda-\lambda^2}{2+a\lambda-\lambda^2}$, $a = c/\sqrt{2}$; $= -\frac{1}{\pi-1}$ at $c=0$
Omitted-variable bias	$b_{Y\cdot X} = \beta + \gamma\,\mathrm{Cov}(X,Z)/\mathrm{Var}X$
Selection-bias decomposition	$\mathbb{E}[Y \mid T{=}1] - \mathbb{E}[Y \mid T{=}0] = \mathrm{ATT} + \mathbb{E}[Y(0) \mid T{=}1] - \mathbb{E}[Y(0) \mid T{=}0]$
Randomized difference in means	unbiased for ATE, $\mathrm{Var} = \sigma_1^2/n_1 + \sigma_0^2/n_0$

THINK ABOUT IT **Berkeley's within-department analysis assumes department choice is a legitimate pathway. If departments with many women were under-funded because they attracted women, which comparison measures discrimination?**

THINK ABOUT IT **An agent's memory stores only trajectories that the user rated highly. Which arrows does the rating add to the agent's causal graph, and what does conditioning on it do?**

THINK ABOUT IT **In a randomized A/B test of two agent versions, 30% of sessions abort before the outcome is recorded, more often for version B. Is the comparison still unbiased?**

Further reading. Pearl, Glymour & Jewell (2016), *Causal Inference in Statistics: A Primer*, Ch. 1–2 (Simpson's paradox, causal graphs); Pearl (2009), *Causality*, Ch. 1 and Sec. 6.1; Hernán & Robins (2020), *Causal Inference: What If*, Ch. 1–8 (confounding, selection bias, M-bias, measurement

Table 12.8. Common pitfalls.

Pitfall	Remedy
Reading $P(y \mid x)$ as an effect	ask which rung the question is on; draw the graph
"Control for everything"	adjust for confounders only; never for colliders, mediators (total effect) or their descendants
Pooled vs. stratified: pick the bigger sample	the choice is causal: stratify over confounders, not over consequences of X
Analysing only completed / surviving units	model the selection $P(S = 1 \mid \cdot)$; selection is not fixed by randomization
Adjusting for a noisy proxy and declaring victory	expect residual confounding; do sensitivity analysis (Session 14)

error); Peters, Janzing & Schölkopf (2017), *Elements of Causal Inference*, Ch. 1–2; Bickel, Hammel & O'Connell (1975), *Science* 187:398–404.

APPENDIX

12.17 Appendix: Proofs

Proof of Lemma 12.1. Let $U \sim N(0,1)$ and $\bar{\Phi}(a) = 1 - \Phi(a) > 0$. Since $\phi'(u) = -u\phi(u)$,

$$\int_a^\infty u\,\phi(u)\,du = \big[-\phi(u)\big]_a^\infty = \phi(a), \qquad \text{so} \qquad \mathbb{E}[U \mid U > a] = \frac{\phi(a)}{\bar{\Phi}(a)} = \lambda(a).$$

Integrating by parts with $u \cdot (u\phi(u))$,

$$\int_a^\infty u^2\phi(u)\,du = \big[-u\phi(u)\big]_a^\infty + \int_a^\infty \phi(u)\,du = a\phi(a) + \bar{\Phi}(a),$$

so $\mathbb{E}[U^2 \mid U > a] = 1 + a\lambda(a)$ and $\mathrm{Var}(U \mid U > a) = 1 + a\lambda - \lambda^2$. Positivity: the conditional law is non-degenerate, so its variance is positive. Upper bound: $\lambda(a) = \mathbb{E}[U \mid U > a] > a$ because $U > a$ on the conditioning event, and $\lambda > 0$; hence $a\lambda - \lambda^2 = -\lambda(\lambda - a) < 0$ and the variance is below 1. Thus $t = a\lambda - \lambda^2 \in (-1, 0)$, which is what Theorem 12.2 uses. □

Simpson's paradox for conditional means. Theorem 12.1 only used $\mathbb{E}[Y \mid X = x] = \sum_z P(z \mid x)\mathbb{E}[Y \mid x, z]$; replacing p_{xz} by $\mu_{xz} = \mathbb{E}[Y \mid X = x, Z = z]$ gives the same decomposition for any outcome with finite mean (Worked Example 12.15). For continuous Z the sums become integrals against $P(dz \mid x)$ and the proof is unchanged.

CONNECTIONS AND TAKEAWAYS

This session deliberately put the data first and the formalism second: the point was to make the insufficiency of $P(Y \mid X = x)$ felt, with real numbers, before we spend three sessions building the machinery to fix it. The theorems we did prove all quantify that same gap. Three things carry forward. First, the vocabulary: confounder (a fork creating spurious

X–Y association) versus collider (whose naive conditioning creates the opposite mistake) — both graphical concepts already available since Session 6, now understood as double-edged. Second, the diagnosis: $P(y \mid x) \neq P(y \mid \text{do}(x))$ in general, and no amount of more data or better inference algorithms closes this gap by itself. Third, and most important: the missing ingredient is not a new independence-reading rule, but a new kind of arrow — one that means "generates" rather than merely "factorizes with." **Session 13** builds exactly that: the structural causal model, in which the DAGs we have been drawing since Session 6 are reinterpreted as literal generative mechanisms, robust enough to support the question "what if this piece were overridden?" that observation alone can never answer.